

Evaluating Explanation Methods for the Classification of Land Use and Land Cover

Student Name: Luca Huelle

Supervisor Name: Paolo Remagnino

Submitted as part of the degree of BSc Natural Sciences to the
Board of Examiners in the Department of Computer Sciences, Durham University

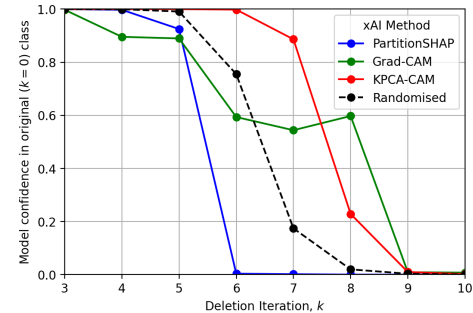
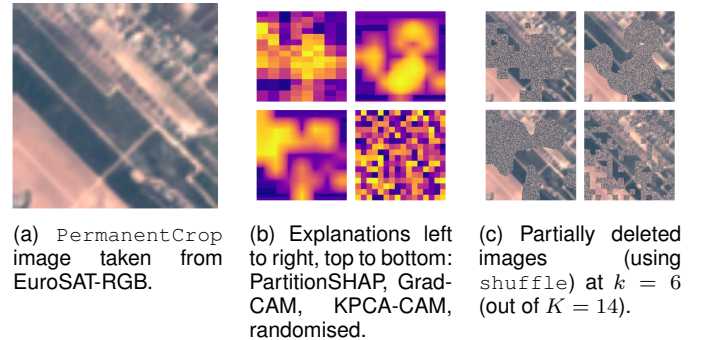
Abstract—We discuss the land use and land cover classification problem for remote sensing, training each of three deep learning architectures on four commonly studied datasets, including on multi-spectral data. While achieving performance en par with the state of the art, we discuss the need for interpretable outputs and hence apply three popular Explainable AI (xAI) techniques, including SHAP and class activation map (CAM) variants, to each of our twelve trained models. Directly addressing the lack of quantitative evaluation of xAI techniques within the remote sensing domain, and the deep learning field more generally, for each xAI method we investigate four properties of the Co-12 framework proposed by Nauta et al. [1] in CORRECTNESS, OUTPUT-COMPLETENESS, CONTRASTIVITY, and COMPACTNESS via six unique metrics, some for the first time to our knowledge in the remote sensing domain. We find that there are no shortcuts for explainability, with some explanations that look convincing, particularly via CAM methods, in fact highly misrepresentative of actual model behaviour, especially for land cover images without a clearly defined object. We also highlight how the performance of xAI methods can vary wildly between dataset and model combinations, underscoring the lack of a ‘one size fits all’ approach to explainability and the need, even beyond the remote sensing domain, to not just study how we develop and train increasingly complex models, but how we build explanations to interpret them too.

Index Terms—Deep learning, Explainable AI, Performance evaluation, Remote sensing

1 INTRODUCTION

TENS OF thousands of kilometres above the earth’s surface, tens of thousands of satellites are in orbit [2], beaming back petabytes of data daily [3] about the surface of our planet. Among these data lie incredibly valuable insights that already inform many areas of policy and decision making on domestic, national as well as international levels in areas across agriculture, natural hazard tracking, urban planning, and monitoring the impacts of climate change. Against the backdrop of the growing significance of all these issues as the global population rises, remote sensing has thus become a critically important field for tracking both natural and anthropogenic change on our home planet.

However, satellite and aerial imaging data such as the example in Figure 1(a), and in particular the task of Land Use and Land Cover (LULC) classification which will constitute the focus for the remainder of this work, presents a unique set of challenges compared to typical imaging data. The first of these stems from the very essence of our planet itself: its surface is incredibly geographically varied and diverse meaning that images, unlike those in classical classification datasets, rarely conform to a single-object distribution. Instead, data have a complex multi-object structure such as an image tile of a motorway also containing smaller roads, junctions, and adjacent fields. Not only is there this “coexistence” [4] of objects within the same image, but these objects can appear at hugely varying scales, contributing to the problem of *intra-class* diversity. Shared features between images, the notion of *inter-class* similarity, compound this issue. Some land cover images may not even contain a primary ‘object’. The second challenge arises from the process of data collection: satellite



(d) Incremental deletion curve, $f_c(\mathbf{x}'(\mathcal{R}, k))$.

Fig. 1. The process of evaluating the CORRECTNESS of three xAI methods via the incremental deletion curve, described in Section 3.4.3. A smaller area under the curve (AUC) is better. PartitionSHAP has an AUC of 5.43 compared to the AUC of a randomised explanation of 6.44. Grad-CAM and KPCA-CAM are both worse than random for this example.

data is composed of numerous multi-spectral (MS) bands corresponding to absorbency/reflectance data of differing surfaces for different wavelengths of light. Modern satellites such as Sentinel 2A, launched as part of the European Space Agency’s (ESA) Copernicus program, have as many as thirteen different bands at different spatial resolutions. Among them are the classical red, green and blue wavelength bands of typical three-channel imagery but also further bands to aid post-processing tasks such as cloud detection [5] and rudimentary LULC segmentation, the latter often taking advantage of band combinations such as the Normalised Difference Vegetation Index (NDVI) to determine areas of dense vegetation for instance.

Together, these challenges naturally posit the computer science question of how we can efficiently unlock insights from this overwhelming quantity of data. Naturally, from the launch of the very first satellites in the 1950s, computational techniques were desired to process these large amounts of data. Originally, as in many other computer vision domains at the time, these took the form of hand-crafted feature-extraction techniques (discussed in more detail in Section 2.1). The development of Convolutional Neural Networks (CNNs) and the resulting explosion of deep learning approaches in the 2010s, arguably initiated by the success of AlexNet [6] in the ILSVRC-2012 image classification challenge [7] (the source of the dataset hereon referred to as ImageNet), hugely accelerated progress on computer vision tasks like classification, naturally extending to domains such as remote sensing.

While these methods have been unquestionably successful in achieving very high accuracies across benchmark remote sensing tasks, their fundamental ‘black box’ mode of operation, where very little or no reasoning is provided by the model to explain its outputs, presents a problem to policy makers and a highly-scientific field which is historically accustomed to a more physically grounded approach. This explainability problem therefore continues to restrict models’ use [8], as well as the development of new techniques, since it remains unclear which aspects of the input the model is grounding its outputs on. Recent legislation, such as the EU’s General Data Protection Regulation (GDPR), has also enshrined the legal right to an explanation for users impacted by technological decisions. The problem is intensified by the unique characteristics of remote sensing presented above which result in complex, multi-faceted images, meaning that when a model may provide only a single output label (or at best, a set of class confidences), much of the underlying detail and nuance of the original images is lost.

In an effort to address these issues present across domains well beyond remote sensing, the deep learning field began to develop Interpretable/Explainable AI (xAI) techniques, aiming to provide stakeholders across a model’s lifecycle development, from system developers to end users, insights into the underlying decision-making process. However, there is a documented lack of quantitative evaluation of these xAI techniques [1], [9], with authors often relying solely on anecdotal evidence and cherry-picked visual examples to supposedly demonstrate the efficacy of their selected methods. This has made it incredibly difficult to concretely compare approaches and to identify aspects

of the explanations which need to be improved, hindering their development and general adoption. Additionally, given that they are often developed for classical computer vision datasets [9], e.g. ImageNet [7], their relevance or ‘fit’ to a specific field such as remote sensing is hard to quantify. This is what motivated Nauta et al. [1] to propose a metrics framework in the form of their “Co-12 properties” which provides a comprehensive set of evaluation criteria that can be applied to all xAI methods.

1.1 Contribution

In this work we investigate four properties of this Co-12 framework (CORRECTNESS, OUTPUT-COMPLETENESS, CONTRASTIVITY, and COMPACTNESS) via six distinct quantitative metrics to evaluate three xAI methods (PartitionSHAP [10], Grad-CAM [11], and KPCA-CAM [12]) on the LULC problem in the remote sensing domain, some for the first time to the best of our knowledge. Specifically, we apply and evaluate these xAI methods on three foundational deep learning architectures in ResNet-50 [13], ConvNeXt [14], and the Swin Transformer [15], trained to performance on par with the state-of-the-art on four well established LULC datasets: EuroSAT [16] (in both its RGB and MS variants), UC Merced [17], and PatternNet [18]. We are thus able to robustly and quantitatively examine the effectiveness of the three selected xAI methods, in answer to the following research question:

How well can xAI techniques be applied to and evaluated on deep learning models for remote sensing LULC classification?

By comparing the xAI methods’ performance when applied across different model and design paradigms, we are able to examine the generalisation ability of these methods to new architectures beyond their original scope and their flexibility across domains more widely, even beyond remote sensing. We also provide the full training and evaluation source code and results of our experiments at <https://github.com/tameTNT/evaluating-xai-remote-sensing>, establishing an easily extensible and modular framework for future work to implement additional evaluation metrics and compute results for further model and dataset combinations within remote sensing and further afield.

2 RELATED WORK

A summary of some of the key works described in this section is presented in Figure 2.

2.1 Deep Learning for Computer Vision and Remote Sensing

The timeline of developments within the remote sensing domain is in many ways a reflection of the progression of the computer vision and machine learning domains. As briefly described in Section 1, before the advent of deep learning techniques, analysis of remote sensing images was mostly achieved by computing hand-engineered image features [4] such as via the Scale Invariant Feature Transform (SIFT) [25], [26] and utilising colour and texture-based descriptors [27]. For scene-level classification tasks, feature encoding and aggregation techniques such as Bag of Visual Words (BoVW)

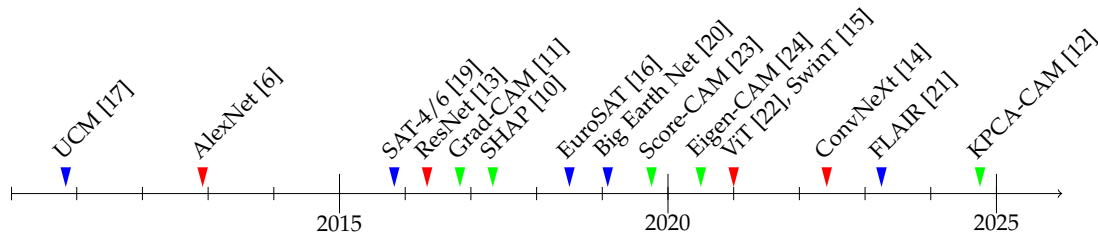


Fig. 2. A timeline of seminal works discussed in Sections 2 to 3. Blue points denote datasets, red points denote deep learning models, and green points denote xAI methods.

were commonly employed, notably by Yang and Newsam [17] who also introduced the small (100 images per category) “UC Merced” dataset, still widely used today and in this work. Neural Network (NN) based approaches, primarily the Multi-Layer Perceptron (MLP), had been discussed and their potential recognised, such as by [28], in the sense that they did not rely as heavily on human model assumptions but instead on their input data. An extension of this idea can be seen nowadays in the success of transformer-based architectures such as the Vision Transformer (ViT) [22] and Swin Transformer [15] which make very few assumptions about the underlying data compared to their predecessors.

First conceptualised in the 1980s [29], Convolutional Neural Networks (CNNs) were designed to act as spatially invariant feature extractors by utilising a kernel that moves progressively across an image to capture feature information. Despite their longstanding basis, limited computational power meant that they did not really become prominent within the deep learning field until the mid 2010s when the efficient utilisation of Graphical Processing Units (GPUs) to speed up the required matrix computations, most notably by AlexNet [6] for the ImageNet competition [7], finally allowed them to realise their potential. CNNs largely consist first of several layers of convolution kernels, differing from their hand-crafted feature-extraction counterparts by the fact that the kernel weights are ‘learned’ during training through the process of backpropagation. This enables them to adapt far more precisely to the input data but is also the reason behind their demand for huge quantities of it during training. The initial AlexNet used a combination of convolutional and pooling layers (used to reduce image dimensions) together with linear/“fully connected” layers, similar to the original MLPs, to perform the final classification. Building on AlexNet’s success, the following years saw the ImageNet competition and the wider research field saw a huge array of models which proposed methods to improve training consistency and overall classification accuracy. GoogLeNet [30] demonstrated the effectiveness of a “deep” architecture, totalling twenty-two layers via the use of “Inception” modules that performed dimensionality reduction before applying convolutions. ResNet [13] went on to demonstrate the training of even deeper networks, with ‘ResNet-152’ having 152 layers, by harnessing residual connections, hypothesised to help with gradient flow during backpropagation which enable the network to more easily represent a wider class of functions.

CNNs’ success in the wider computer vision field inevitably spurred further research into their specific feasibility for remote sensing imagery, as discussed by [31].

Progress was initially limited however by the lack of large-scale (that is, on the same order as ImageNet’s 14 million images) datasets specific to the field [32], so efforts were concentrated on transfer learning and fine tuning in order to achieve strong results. Transfer learning refers to the process of taking a model that has been ‘pre-trained’ on a different image domain and training (part of) it further with new data to improve accuracy in a new domain, such as satellite imagery, under the assumption that certain underlying characteristics about image features are transferable between domains. Together with data augmentation techniques (creating new data by applying transformations such as rotations and flips to existing data), this reduces the amount of raw domain-specific data, which may be difficult to acquire, required. While transfer learning techniques enabled significant progress to be made in the LULC classification domain by e.g. [33], the eventual release of several larger remote sensing datasets for the task, including SAT-4/6 [19], EuroSAT [16], Big Earth Net [20], and SEN12MS [34], the latter three of which also included MS components, spurred further success in developing classifiers trained specifically for the remote sensing domain. Big Earth Net also notably included low resolution segmentation maps and multi-class labels enabling the training of more complex semantic segmentation models. The recently introduced FLAIR dataset [21] aims to push the development of such models further by providing even higher resolution (20 cm) pixel annotated aerial images alongside geo-referenced, multi-temporal satellite imagery. Further work by [35] standardising the format and preprocessing of many of these datasets further aided progress towards high accuracies, upwards of 99% on EuroSAT for instance¹, and reliable models.

As catalogued by [4], NN development for the remote sensing domain is by no means limited to CNNs. Jozdani et al. [36] describe how Generative Adversarial Networks (GANs) for instance, originally proposed by [37], have been trained so as to use the discriminator model as a semi-supervised feature extractor [38], [39]. But more recent research, e.g. [40], has naturally focused more heavily on the transformer architecture [41], given its success in the natural image domain by utilising multiheaded attention in place of CNNs’ convolutions to capture inter-image contextual relationships.

1. <https://paperswithcode.com/sota/image-classification-on-eurosat>

2.2 Explainable AI for Image Classification and Remote Sensing

The eventual success of CNNs for classification tasks drew increasing attention to the ‘black-box’ nature of their outputs: the sense that these models provided little to no reasoning for their decisions. This quality severely limited the ability of stakeholders to assess trust [42], especially in situations where real-world actions were to be taken based on model outputs. At first, analysis of AI outputs was performed in a more general sense, using platforms such as Gestalt [43] and ModelTracker [44] to highlight similar data instances in the training and test domains that could then be used to infer and attempt to generalise model behaviour and learnt ‘rules’. However, as models grew more complex, more in-depth methods to explain specific models’ reasoning were developed; these more targeted approaches fall under the umbrella of xAI. Focusing on the remote sensing domain, Höhl et al. [9] describe the variety of purposes for which such xAI explanations are desired in the first place: *to control* models’ behaviour, for example to adapt to practical out of domain contexts; *to discover* previously unknown characteristics or biases in the input data, thereby informing future feature extraction and deep learning methods; *to improve* models’ overall performance by providing a further avenue for evaluation and to validate that expected image regions form the basis for outputs; and *to justify* their outputs in decision-critical contexts requiring accountability such as the allocation of agricultural subsidies.

Returning to an examination of the methods themselves, building on the work of [45] presenting their ‘deconvolution’ method to map activations in later layers of a ConvNet model back to the input space, Simonyan et al. [46] initially proposed two ways of visualising the outputs of this same model: a “class model visualisation” depicting the amalgamated representation of the class itself within the model, calculated via a maximisation of the class’s confidence score, and a saliency map of the “spatial support” of a class in an image by approximating the classifying function to linear order and determining the pixels of greatest importance. Zeiler and Fergus [45] also suggested iteratively occluding different parts of the image and examining the impact on the output class confidences as a way to highlight the model’s primary image regions of interest. While these methods were specific to image classification or even to a particular model, Ribeiro et al. [42] proposed Local Interpretable Model-agnostic Explanations (LIME) which can be applied to any classification task by approximating the classifier function with a simpler, most commonly linear, function around the decision boundary, learned by feeding perturbed samples into the model. In the case of explaining image classification, perturbations take the form of blurring or removing super-pixels (contiguous groups of similar pixels), determined by classical segmentation algorithms such as Quickshift [47]. Verma et al. [48] demonstrated the LIME technique on their own custom CNN architecture trained on EuroSAT remote sensing data, using it to reveal proxies in the input data that their model had learnt instead of the expected regions of interest for the highway class for example.

Rather than generating explanations via modifying input

images to the model, a range of xAI methods operating more directly on backpropagating activations through the network have also been proposed, including Deep Learning Important Features (DeepLIFT) [49], which describes the difference in a model’s outputs in terms of the difference in input from a baseline ‘reference’ input (a black, blurred or even random noise), and Integrated Gradients (IG) [50], which sums (formally as a path integral) the output gradients of the model along an interpolation from a similar ‘reference’ input to the actual input. While these methods have desirable theoretical characteristics, that is, they are constructed to obey certain “axioms”, for image tasks the selection of an appropriate baseline input is challenging, since it can have large impact on generated output [51], and exact calculation can be computationally expensive. Lundberg and Lee [10] presented a unifying framework for these methods in the form of SHapley Additive exPlanations (SHAP), grounded in game theoretic Shapely values [52], which [53] used to explain the predictions of models trained on the EuroSAT dataset. Class Activation Mapping (CAM) based methods, originally proposed by [54], and their variations in the form of Grad-CAM methods [11], the latter of which computes the gradient of a particular class score with respect to feature map activations of the chosen convolutional layer (discussed in more detail in Section 3), are particularly popular in classification tasks and, since they rely only on a single output (and/or its gradients), are much more computationally tractable. Sen and Keles [55] for instance, use Grad-CAM explanations to demonstrate the learned semantic remote sensing features of their proposed architectures that enable them to generalise outside of their training dataset without additional fine tuning. The popularity and ease of use across models of such CAM methods—including adaptations to more recent transformer architectures such as by [56] and applied to the EuroSAT dataset by [57]—has encouraged the development of many variations and extensions including Grad-CAM++ [58], Score-CAM [23], Eigen-CAM [24], and Channel Saliency and Gradient (CSG) CAM [59], the last of these specifically for the remote sensing domain.

However, the outputs of these xAI methods are rarely scientifically and repeatably compared nor their validity quantitatively demonstrated, with authors instead relying on anecdotal visual evidence in many cases and rarely numerically quantifying improvements [1]. This means that proposed xAI methods in some cases may not be interpreting the model’s output at all but merely acting as edge detectors on the input, simply reflecting the underlying model architecture as demonstrated in [60]. Some proposals do evaluate their methods directly via user studies—comprehensively reviewed by [61]—but these results are also difficult to compare and replicate while also being very time and resource intensive. Nauta et al. [1] therefore review and collate the already proposed quantitative evaluation metrics throughout the literature, categorising them based on which aspect of explanation they evaluate through their proposed “Co-12 properties”; Table 2 of [1] in particular provides a succinct summary for the foundation behind each of the twelve properties and further stratifies them into properties evaluating an explanation’s suitability for the user, its presentation, and of course the content of

the explanation itself. The four specific properties examined by this work are described in detail in Section 3.4.3.

Höhl et al. [9] review the use of xAI techniques within the remote sensing field specifically, highlighting the popularity of SHAP-based techniques generally in the LULC mapping task and CAM-variants for CNN models. In accordance with the findings of [1] for the general xAI field, they also find that the majority of authors evaluate their use of xAI methods only anecdotally; SHAP explanations in particular are almost never quantitatively evaluated. Kato-georgiou and Karantzas [62] evaluate ten different xAI methods on two CNNs in DenseNet-121 [63] and ResNet-50 [13] trained on the Big Earth Net [20] and SEN12MS [34] datasets but ultimately only quantitatively examine three of the Co-12 properties (metric classification ours): they examine CORRECTNESS by calculating the area under the most relevant first incremental deletion curve [64]; CONTINUITY via the max sensitivity metric [65]; and COMPACTNESS via the proxy of file size [66]. In their evaluations, they focus primarily on the explanations’ adherence to domain expectations in terms of identifying the anticipated regions of interest, which [1] explicitly cautions against, favouring instead an evaluation of the explanation’s faithfulness to the underlying model itself.

3 METHODOLOGY

This section presents the three distinct stages of our approach beginning first with the training of the three selected deep learning models on the four nominated datasets before moving on to an in-depth discussion of the considered xAI methods and concluding with an explanation of the quantitative evaluation criteria applied to them.

3.1 Selected Models and Training

In the case of all three deep learning architectures, we use the implementations provided by the `torchvision` package [67], modifying the first input/final linear layer in each case to match the number of input channels/output classes required and, for the RGB datasets, fine tune the provided weights from pretraining on the ImageNet dataset [68] of the final output layer using a higher learning rate, α_1 , before unfreezing and training the full network with a lower learning rate, α_2 , thereby following the general regime presented by [16]. We train to minimise the cross-entropy loss using the AdamW optimiser [69] alongside a learning rate scheduler, `ReduceLROnPlateau`, to gradually decrease the learning rate when the validation loss plateaus, that is, does not improve by more than threshold v for five epochs, following [70]. We also utilise early stopping once the learning rate has been successively reduced eight times to save computation resources.

ResNet-50 [13] is selected due to its prevalence throughout the literature and well-established role as a benchmark classical CNN model for remote sensing LULC tasks as discussed in Section 2, including in the original datapapers [16], [18]. The architecture is primarily made up of convolutional blocks and pooling layers which progressively reduce the spatial dimension, while easing data flow by notably introducing residual connections between layers. The model

has approximately 23.5 million parameters (after adjusting input and output layers).

ConvNeXt [14] is selected as a modern incarnation of the convolution-based architecture, originally presented as a refinement “for the 2020s” of the aforementioned ResNet architecture, in particular via making use of the priors developed for vision transformers [22] alongside depth-wise convolutions, inverted bottlenecks and larger kernel sizes (7×7 versus ResNet’s primarily 3×3 kernels). We use the Small variant with approximately 49.5 million parameters largely due to computation limitations. By selecting this model we wish to investigate: (a) whether the xAI methods we choose, most designed during the dominance of ResNet-like architectures, are *robust and flexible* enough to explain modern combinations of existing building blocks; and (b) whether the methods can still reliably explain deeper networks with a much larger number of parameters.

Lastly, we examine a purely transformer-based architecture in the **Swin (Shifted Window) Transformer (SwinT)** [15] selected for two reasons: firstly, because of its role as the inspiration for the design decisions of the ConvNeXt architecture discussed above, and secondly due to its demonstrated effectiveness over the baseline ViT architecture [22] specifically for remote sensing tasks [71]. Again, we use the Small (v2 [72]) variant with approximately 49.0 million parameters. By including such a transformer-based model, we aim to explore whether the xAI methods *generalise* to new design paradigms outside their original domain, having emerged well after their original development.

3.2 Datasets and transforms

We train our selected models on four main datasets, each split into a training, validation and test set (in the ratio 60/20/20% respectively), taking advantage of the `torchgeo` package [73]. All results and visualisations presented in this paper use the test set. Before feeding the images into the model, we resize them to match the expected model input dimensions (224×224 for ResNet-50 and ConvNeXt and 256×256 for SwinT) using bilinear interpolation. We apply additional transformations depending on the dataset as detailed below. Collated examples from each dataset are shown in Figure 3.

Firstly, we include **UC Merced (UCM)** [17] as a seminal high-resolution (about 30cm/pixel) remote sensing dataset covering 21 different LULC classes. Each image is 256×256 pixels collected from aircraft observations extracted from the USGS National Map Urban Area Imagery collection. With just 100 images per class and a mix of land *cover* and land *object* scenes designed to often share textures/features and classes that frequently overlap (e.g. separate sparse/medium/dense residential classes), this dataset can be challenging. However, its high resolution means that its images are likely to share many of their low-level features with those of the traditional image datasets that models were pretrained on, making it highly suited for fine-tuning [33]. We follow the transformation regime described by [32]—namely, randomly taking the transpose of images, rotating a multiple of 90 degrees, and rotating up to a further 7 degrees—who motivate their proposals by noting that remote sensing imagery’s viewpoint high

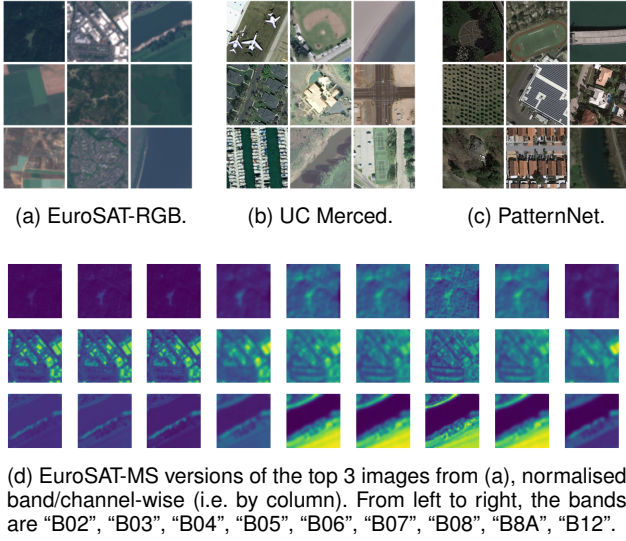


Fig. 3. Image samples from each dataset. Notice how UC Merced and PatternNet have a higher spatial resolution than EuroSAT and how similar classes, such as rivers, are present across all datasets.



Fig. 4. Selected images from EuroSAT-RGB. The *Highway* images in the top row have high *intra-class* variation with different orientations, sizes and surroundings, while sharing many features with the *Pasture* and *Residential* images shown in the bottom row (high *inter-class* similarity).

above captured objects means the images have no natural ‘upright’ position. This implies that models should be made to learn classification “without regard to feature orientation” via the use of image augmentations such as those suggested. This is especially important for very small datasets, such as UCM, which might have an unintentional embedded positional/rotational bias. We scale the input data from the range $[0, 255]$ to the range $[0, 1]$ (via $x_{out} = (x_{in} - x_{min}) / (x_{max} - x_{min})$) before normalising assuming mean and standard deviation of 0.5 to provide unit Gaussian input data, $x \sim \mathcal{N}(0, 1)$, to the model in the final range $[-1, 1]$.

As an example of MS satellite imagery, we investigate **EuroSAT** in both its RGB and MS variants [16]. We refer to these as *EuroSAT-RGB*² and *EuroSAT-MS* throughout the rest of this work. The dataset contains 27,000 Sentinel-2A 64×64 pixel image patches, each with a corresponding LULC label from one of ten classes with the goals of: attaining high variance by including data over the course of an entire year; directly selecting to maximise intra-class variation and inter-class similarity (as discussed in Section 1 and

demonstrated in Figure 4), so as to discourage learned dependence on low-level features; and discriminating within generalised classes such as agricultural land use (split into *AnnualCrop*, *PermanentCrop*, and *Pasture*), built-up areas, and bodies of water. For *EuroSAT-RGB*, we follow the regime of the original paper, projecting values from the reflectance value input range $[0, 2750]$ to the image range $[0, 1]$ before normalising again as described above for UCM. For *EuroSAT-MS*, we first select a nine-band subset (see Figure 3(d)) of the full thirteen bands available, given that bands “B01”, “B09”, “B10”, and “B11” are collected mainly for the purposes of atmospheric corrections and preprocessing (e.g. filtering out high cloud-cover images, which has already been performed by the creators of *EuroSAT*) and not LULC observations, similarly to [74], [75]. To normalise the data to be approximately normally distributed and thereby ensure convergence during training, we utilise the mean and standard deviation calculated across the entire training dataset to directly scale the data starting from the originally provided reflectance values. For both variants, we additionally apply several random transformations—including horizontal flipping, and affine transformations of rotation (up to 2 degrees anti-/clockwise), translation (by up to a tenth of each images’ height/width along both axes), scaling (0.95-1.05 of the original image size), and shear (2 degrees in either direction along the x-axis)—in order to augment the training data and avoid overfitting.

Finally, we train on the **PatternNet** dataset [18], originally designed for the task of remote sensing image retrieval (RSIR). Similarly to UCM, this dataset is made up of 256×256 pixel images of 38 different LULC and object classes, such as golf course, shipping yard, dense residential, and river, collected from Google Earth imagery. With 800 images per class and a higher number of classes overall, this dataset was created specifically for deep learning applications which require more training data than the hand-crafted feature methods discussed in Section 2 and used by UCM’s authors. Notably, the surface-level resolution of the images also varies from 0.062 to 4.693 m/pixel so as to minimise background information around the class of interest. We apply the same augmentations and normalisation procedure as with UCM to PatternNet.

3.3 xAI Methods

This work examines three different post-hoc (explaining an already trained predictive model) xAI methods which generate heatmaps, selecting SHAP and two CAM-based methods given their established popularity in the remote sensing community [9].

3.3.1 SHAP

We use the *shap* package to calculate pixel-level SHapley Additive exPlanations (SHAP) [10]. This yields an approximation of Shapley values from coalitional game theory [52], formally defined in terms of feature index subsets $S \subseteq F$ as:

$$\phi_q = \sum_{S \subseteq F \setminus \{q\}} \frac{|S|!(|F| - |S| - 1)!}{F!} [f(x_{S \cup \{q\}}) - f(x_S)]. \quad (1)$$

Each of these ϕ_q is an importance value representing the positive or negative contribution of the inclusion of that

2. The Red, Green, and Blue (RGB) wavelengths correspond to bands “B04”, “B03” and “B02” respectively [5].

‘feature’ (an individual image pixel), x_q , to the final output/class confidence, f (also referred to in the game theory literature as the “payout”). In other words: how much did a particular feature contribute to the prediction compared to the average prediction [76]? Calculating this exactly requires summing over all possible feature coalitions (subsets S), i.e. over the power set, $\mathcal{P}(F)$, so has exponential runtime, 2^Q (where $Q = hw$ is the number of pixels in an image of height h and width w , each either included or not included in the subset).

Using Shapley values to explain machine learning model output had been proposed before, but Lundberg and Lee [10] unified the values’ calculation with several other explanation methods, including LIME [42] and DeepLIFT [49], classing them all as “additive attribution methods” of the form,

$$g(z') = \phi_0 + \sum_{q=1}^Q \phi_q z'_q, \quad (2)$$

where the explanation model’s output, g , for simplified input (coalition vector), $z' = \{0, 1\}^Q$, specifying which features, x_q , are present, receives a contribution $\phi_q \in \mathbb{R}$ from each feature; in our case, these ϕ_q are the Shapley values. Explanation models of this form fulfil several desirable properties, namely: local accuracy, missingness, and consistency. For more detail, the reader is referred to [10], [76].

To enable efficient computation in a model-agnostic way, we use the PartitionSHAP method³ which approximates the Shapley values recursively using a “hierarchy of features” and Owen values. Since, in the case of explaining an image, a particular feature coalition refers to an image where some regions (i.e. the features not in the coalition currently being considered) are masked (we mask via blurring with a 128×128 kernel), the method uses a balanced partition tree of images with blurred regions, iteratively including/excluding smaller and smaller parts of the image⁴. The beginning of this process is visualised in Figure 5. By prioritising larger areas/coalitions first before moving on to smaller and smaller regions, we are also able to interrupt the calculation before considering the full partition tree—we impose a limit of $l = 3000$ forward passes through the model being explained per original input image—to minimise the effect of the otherwise still quadratic runtime in Q while maintaining acceptable resolution and accuracy of the final output.

3.3.2 Grad-CAM

Incredibly commonly used for CNNs, Gradient Class Activation Maps (Grad-CAM) [11] are a fast way (requiring only one forward and backward pass through the model being explained) to directly generate explanations based on network activations within a convolution-based model. We apply the following process to the final convolutional layer (with n output channels/feature maps) of each model, believed to capture the highest-level concepts (with the largest receptive field) required for ultimate classification while still

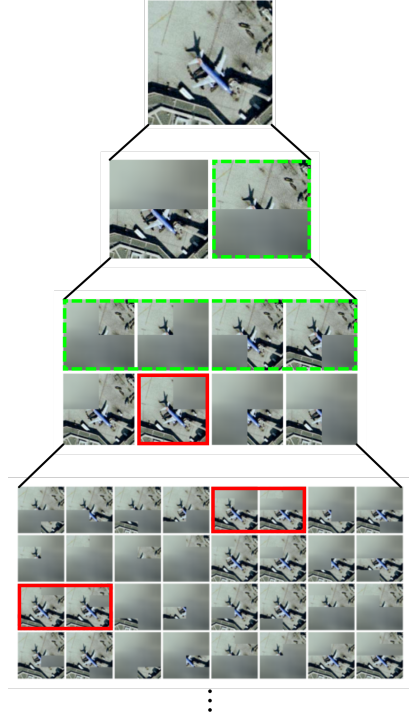


Fig. 5. Chronologically from top to bottom, an illustration of the masked images passed to the model by the PartitionSHAP function using an airplane image from UCM. The original image is masked in successively smaller regions with two regions added and two subtracted from each image each stage as highlighted by the dashed green border moving from the stage 2 to 3 and then by the red border moving from stage 3 to 4. This process continues until the total number of images the model has been evaluated on reaches l .

retaining the spatial information lost in later linear layers. For SwinT, we extract the values after the final normalisation layer and before average pooling⁵, reshaping them to the same dimensions ($n \times u \times v$) as a typical convolution layer. Since the dimensions of the CAM, $L \in \mathbb{R}^{u \times v}$, are by default of a very low resolution, the same as those of the convolutional feature map itself (e.g. 7×7 for ResNet-50), we upscale them to match the input image, $x \in \mathbb{R}^{h \times w}$, utilising the implementation provided by the `pytorch-grad-cam` package [77].

As visualised in Figure 6, for the given score (the output logit before softmax is applied) of a particular classification class c , y^c , Grad-CAM weights each $u \times v$ feature map, $A^k \in \mathbb{R}^{u \times v}$, of the selected convolution layer by a factor α_k^c in a linear combination, retaining only positive influences on the final output class by applying the ReLU function ($\text{ReLU}(x) = \max(0, x)$):

$$L_{\text{Grad-CAM}}^c = \text{ReLU} \left(\sum_{k=1}^n \alpha_k^c A^k \right). \quad (3)$$

These “neuron importance weights”, α_k^c , are designed to capture the importance of feature map k to the classification y^c via a partial linearisation. They are therefore calculated

5. For the `torchvision` implementation we use, this corresponds to the output of the `Permute` module. See https://github.com/pytorch/vision/blob/95f10a4ec9e43b2c8072ae5a68edd5700f9b1e45/torchvision/models/swin_transformer.py#L596

3. <https://shap.readthedocs.io/en/latest/generated/shap.PartitionExplainer.html>

4. Channels are *not* considered individually.

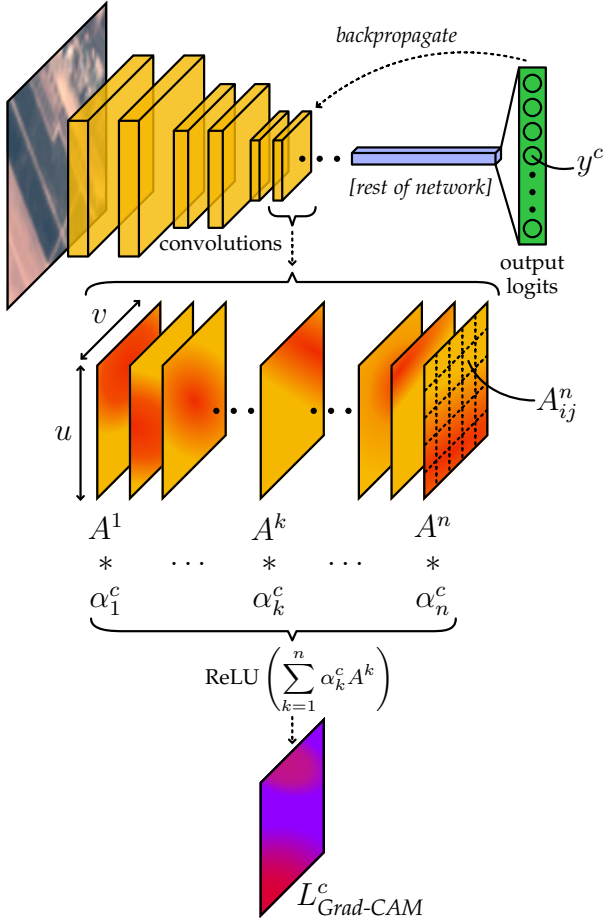


Fig. 6. An annotated diagram describing the Grad-CAM xAI method for a single image input. The α_k^c weights for each feature map A^k of the targeted convolutional layer are given by Equation (4).

via a global average pool (that is, over all units, i, j , of each feature map) of the gradients (calculated via backpropagation) of y^c with respect to each feature map unit A_{ij}^k , as given by:

$$\alpha_k^c = \frac{1}{uv} \sum_{i=1}^u \sum_{j=1}^v \frac{\partial y^c}{\partial A_{ij}^k}. \quad (4)$$

3.3.3 KPCA-CAM

The same package allows us to approach the final xAI method we investigate, KPCA-CAM [12], as drop-in replacement for Grad-CAM. A much more recent variant of CAM designed to capture non-linear relationships within CNN activations more effectively, its authors claim it produces more precise activation maps. KPCA refers to Kernel Principal Component Analysis [78], an extension of principle component analysis (PCA) which first applies a *non-linear* mapping of the input data into a higher dimensional space through the use of a kernel function before applying PCA. The kernel function the original authors show to be most effective is the sigmoid kernel, defined between two $\mathbb{R}^n$ feature vectors, $\mathbf{x}_a$ and $\mathbf{x}_b$, as:

$$k(\mathbf{x}_a, \mathbf{x}_b) = \tanh(\gamma \mathbf{x}_a^\top \mathbf{x}_b + c_0), \quad (5)$$

where γ is the scale/slope parameter (commonly set to $1/n$, as in `scikit-learn`) and c_0 is the intercept/offset (set to 1 by default).

For the case of explaining an input image via KPCA-CAM, there are $u \cdot v$ feature vectors (one per unit of the feature maps) each consisting of n activation values, one for each feature map in the layer. Formally this is given by $\mathbf{x}_{(i-1)u+j} = (A_{ij}^1, \dots, A_{ij}^k, \dots, A_{ij}^n)^\top$, using the same notation as in equation (4). To produce the final CAM, $L_{KPCA-CAM}$, we project the kernel matrix $K \in \mathbb{R}^{uv \times uv}$, defined via $K_{ab} = k(\mathbf{x}_a, \mathbf{x}_b)$, onto its largest (that is, corresponding to the largest eigenvalue) eigenvector, $\mathbf{v}_1 \in \mathbb{R}^{uv}$:

$$L_{KPCA-CAM} = \underset{\mathbb{R}^{uv} \rightarrow \mathbb{R}^{u \times v}}{\text{reshape}} (K \mathbf{v}_1). \quad (6)$$

This equates to each feature vector, $\mathbf{x}_c$, being projected onto the 1st (since we take only the largest eigenvector) principal component, thereby aiming to capture the abstract non-linear features learnt by the CNN inherent in its activations in a way which, the authors argue, Eigen-CAM [24] is unable to.

Note that this technique does not require the model's gradients, merely the activations (feature maps) of the selected convolutional layer from one forward pass on the desired input to be explained. Therefore, it can only be used to explain the model's actual output (unlike Grad-CAM which works for any desired class c) but still remains model-agnostic, requiring only a convolution-like layer. The same upscaling process as for $L_{\text{Grad-CAM}}$ is still needed.

3.4 Evaluation of xAI Methods

Lastly, we discuss the metrics we use to evaluate the effectiveness of each of the xAI methods on our selected remote sensing datasets. As highlighted by [79], precisely how these metrics are calculated can greatly affect their output so detailed parameter selections are given in Section 4.1 and in Table 1. We follow the ‘Co-12 properties’ categorisation framework proposed by [1] and adopt the same notation where f denotes the predictive model being explained and e denotes an explanation method which generates explanations such that $g(\mathbf{x}) = e(f(\mathbf{x}))$ ultimately represents the explanation generated by e for the output of f evaluated on input sample $\mathbf{x}$.

3.4.1 Ranked Explanations

In most cases, metrics require the consideration of ‘important’ pixels: these are determined by converting explanation heatmaps for the output class of f into ‘ranked’ explanations assigning a value of 0 to the pixel in the explanation with the highest value and a value of $Q - 1$ to the pixel with the lowest value (potentially negative). This can be formalised as redefining the output of e , the heatmap matrix H with value $H_{ij} = H(i, j) = H(\mathbf{r}_p)$ at pixel coordinates $\mathbf{r}_p = (i, j)$ in the image, as an ordered (ranked) set, $\mathcal{R}$:

$$\mathcal{R} = (\mathbf{r}_0, \dots, \mathbf{r}_p, \dots, \mathbf{r}_{Q-1}) \quad (7)$$

where $p_1 < p_2 \Leftrightarrow H(\mathbf{r}_{p_1}) > H(\mathbf{r}_{p_2})$ as in [64].

3.4.2 Explanation Similarity

Many metrics also necessitate the calculation of a notion of similarity between generated explanations. To enable comparison between xAI methods which use different output ranges, we only ever compare ranked explanations. Following Adebayo et al. [60], we use the perceptual structural similarity index measure (SSIM) [80] which, as the name suggests, captures structural information more effectively than a more rudimentary mean squared error (MSE) comparison thanks to its incorporation of texture information [81]. 1 indicates the greatest similarity whereas 0 indicates no similarity⁶.

Since we are often only interested in the areas deemed most important by e , we also calculate the top- m intersection, I_m , suggested by [82], between two ranked explanations $\mathcal{R}_1$ and $\mathcal{R}_2$ as

$$I_m = \frac{|(\mathcal{R}_1)_{0:m-1} \cap (\mathcal{R}_2)_{0:m-1}|}{m}, \quad (8)$$

where 1 again indicates the highest possible similarity.

Finally, the ranked nature of explanations lends itself naturally [60], [82], to the calculation of the Spearman rank-order correlation coefficient, ρ , between two explanations, where 0 indicates no correlation (completely dissimilar) and ± 1 indicates positive/negative correlation (exactly the same/inverses of each other).

3.4.3 Evaluating Metrics

We first examine the property of CORRECTNESS which addresses the “descriptive accuracy” [83] of e to f ; in other words, e ’s faithfulness to the true inner workings of the black box f . It is crucial to note that an explanation which looks plausible to a user may not be true to f ’s reasoning. As a basic ‘sanity check’ therefore, we perform a **Model Parameter (Cascading) Randomisation Test** as introduced by [60], fully randomising all model parameters to their PyTorch defaults and calculating the similarity, as described in Section 3.4.2, between the explanations generated for this randomised model and those of the trained model for the same inputs. For a faithful e sensitive to f , we expect low similarity.

This is a necessary but not sufficient condition on the correctness of e , so we also calculate the **Area Under Most Relevant First Incremental Deletion Curve** (AUC-MoRF) metric as proposed by [64], [84], and similarly to the work by [62]. For each iteration k of K total iterations, we ‘delete’ the top $t_k = \lfloor hw \cdot k / K \rfloor$ most relevant pixels from $\mathbf{x}$ (that is, the pixels at coordinates $\mathbf{r}_p \in \mathcal{R}$ such that $p < t_k$) to produce a new input $\mathbf{x}'(\mathcal{R}, k)$ and record the model’s confidence in the originally predicted ($k = 0$) class for the new input, $f_c(\mathbf{x}')$. Notably, there are several methods of deleting pixels/generating $\mathbf{x}'$, visualised in Figure 7⁷. We can then plot the (discrete) *incremental deletion curve*, $f_c(\mathbf{x}'(\mathcal{R}, k))$, over k ,

6. Negative values up to -1 are also possible, indicating anti-similarity (inversion). See this GitHub issue: <https://github.com/jterrace/pyssim/issues/15#issuecomment-216065455>.

7. A common criticism of pixel perturbation/deletion methods is that they can lead to samples outside of f ’s input domain, as observed by [85]. The shape of the deleted region has also been shown to potentially leak class information to the model [86]. However, detailed further analysis is beyond the scope of this work



Fig. 7. The effect of various deletion methods on the same 60% randomly selected pixels of a wastewater treatment plant image from PatternNet. From left to right: the original image, constant replacement, Gaussian noise, blurring, in-painting, fill via nearest neighbour, and pixel shuffling.

as visualised in Figure 1(d), and calculate the area under this curve (AUC)⁸ using the trapezoidal rule:

$$\text{AUC}(f, \mathbf{x}, \mathcal{R}) = \frac{1}{2} \sum_{k=1}^K [f_c(\mathbf{x}'(\mathcal{R}, k-1)) + f_c(\mathbf{x}'(\mathcal{R}, k))] . \quad (9)$$

Following the advice of [1], [62] to view the AUC only in the context of a baseline, we compute the ratio of this area to that of the randomised incremental deletion curve which uses the mean curve $f_c(\mathbf{x}'(\tilde{\mathcal{R}}, k))$ of an ensemble of T randomised ranked explanations⁹, $\tilde{\mathcal{R}}_t$, informing deletion:

$$\text{AUC}_{\text{ratio}} = \frac{\text{AUC}(f, \mathbf{x}, \mathcal{R})}{\frac{1}{T} \sum_{t=1}^T \text{AUC}(f, \mathbf{x}, \tilde{\mathcal{R}}_t)} . \quad (10)$$

For a good explanation, we expect $\text{AUC}_{\text{ratio}} < 1$ since this indicates that e was better than random at identifying pixels truly important to f with a more steeply decreasing deletion curve and hence a lower AUC.

Related to this metric, we also evaluate OUTPUT-COMPLETENESS (OC), which assesses how much of f ’s behaviour e is able to describe, via the **Deletion and Preservation Check** metrics in the vein of [87]. For the deletion check, this is simply calculated as the change in confidence, $\Delta \mathcal{C}$, versus the original input at a particular point on the deletion curve: $\Delta \mathcal{C}_{\text{del}} = f_c(\mathbf{x}'(\mathcal{R}, \mu K)) - f_c(\mathbf{x})$ for some proportion $\mu \in (0, 1)$. For the preservation check, we invert $\mathcal{R} \rightarrow \mathcal{R}^{-1}$ (to delete the least relevant pixels first) and then calculate $\Delta \mathcal{C}_{\text{pres}} = f_c(\mathbf{x}'(\mathcal{R}^{-1}, (1 - \mu)K)) - f_c(\mathbf{x})$ instead. Again, these values should be compared to their randomised versions, $\Delta \tilde{\mathcal{C}}_{\text{del/pres}}$, as was done for the AUC metric:

$$\text{OC}_{\text{del}} = \Delta \tilde{\mathcal{C}}_{\text{del}} - \Delta \mathcal{C}_{\text{del}} \quad (11)$$

$$= \frac{1}{T} \sum_{t=1}^T [f_c(\mathbf{x}'(\tilde{\mathcal{R}}_t, \mu K)) - f_c(\mathbf{x}'(\mathcal{R}, \mu K))] , \quad (12)$$

$$\text{OC}_{\text{pres}} = \Delta \mathcal{C}_{\text{pres}} - \Delta \tilde{\mathcal{C}}_{\text{pres}} . \quad (13)$$

The latter defined so that the best value for both metrics is 1 ($\Delta \mathcal{C}_{\text{del}} \approx -1$ as we deleted important pixels and $\Delta \tilde{\mathcal{C}}_{\text{del}} \approx 0$; $\Delta \mathcal{C}_{\text{pres}} \approx 0$ as we kept important pixels and $\Delta \tilde{\mathcal{C}}_{\text{pres}} \approx -1$) with negative values indicating randomly selecting pixels for deletion/preservation was more effective at dropping/retaining model confidence.

8. Note that some authors, e.g. [64], choose instead to calculate the *decrease* in model confidence and hence the *Area Over* the deletion Curve (AOC).

9. In practice, we generate random ranked explanations, $\tilde{\mathcal{R}}_t$ in the format of a $d \times d$ square grid with each grid element of uniform importance to ensure that nearby pixels are assigned similar rankings, just as they are in non-randomised ones, $\mathcal{R}$. In Figure 7, we use $d = 8$ for example.

TABLE 1
Training and Evaluation Configuration

Area	Parameter Description	Configuration
Training	Frozen fine-tuning learning rate	$\alpha_1 = 1.2 \times 10^{-2}$
	Full network fine-tuning learning rate	$\alpha_2 = 1.2 \times 10^{-3}$
	Learning rate plateau reduction threshold	$v = 0.1$
Similarity	Top- m intersection	$m = \lfloor hw/10 \rfloor$
CORRECTNESS	Incremental deletion iterations	$K = 14$
	Random explanation ensemble size	$T = 5$
	Grid size for random explanations $\mathcal{R}_t$	$d = 16$
OUTPUT-COMPLETENESS	Deletion/Preservation proportion	$\mu = 0.2$
COMPACTNESS	Sparsity threshold	$\tau = 0.5$

If f 's output changes significantly, even changing its predicted class, for perturbed input $\tilde{x}$, we desire an equally significant change in g . We therefore evaluate CONTRASTIVITY as a measure of the discriminativeness of e with respect to f 's output, the aim being to enable comparison to other outputs [1]. Miller [88] asserts that, in line with the ‘‘contrastive explanations’’ given by humans when asked to explain an event, e should not just explain why the output was f but should also explain why the output was not f' . In a very similar fashion to [89], we probe the **Target Sensitivity** of our selected xAI methods by generating an adversarial example [90] of x , $\tilde{x}$ (a virtually imperceptibly perturbed input designed to fool f to produce an entirely different output (predicted class)), and comparing the explanations for the outputs $f(x)$ and $f(\tilde{x})$, g and $\tilde{g}$ respectively. We seek a *large* difference (low similarity) between g and $\tilde{g}$ for a good explanation model e .

Finally, we propose to measure the COMPACTNESS of generated explanations directly (i.e. not via ranked versions) via a simple **Sparsity**, S , calculation in terms of g 's normalised heatmap, $\bar{H}_{ij} = \text{abs}(H_{ij}) / \max_{k,l}(H_{kl})$, parametrised by some threshold, $\tau \in (0, 1)$:

$$S(H, \tau) = \frac{|\{H_{ij} : \bar{H}_{ij} \leq \tau\}|}{|H|}, \quad (14)$$

where $|X|$ denotes the number of elements of a matrix or set, X . S thereby represents the proportion of pixels *not* marked as important (i.e. those having low normalised heatmap values, $\bar{H}_{ij}$), calibrated via τ . Explanations should be sparse (high S) with a minimal number of important pixels to avoid overwhelming users: ‘less is more’ [91], [92].

4 RESULTS

4.1 Experimental Setup

We train our twelve (4 datasets $\times$ 3 models) $\{\text{dataset}, \text{model}\}$ combinations (DMCs) using Python (3.10.6) and PyTorch [93] (2.5.1) for Linux on a 10GB MIG partition of an NVIDIA A100 GPU alongside 28GB of RAM on the NVIDIA Compute Cluster (NCC) provided by the Computer Science

TABLE 2
Trained Model Accuracy

Dataset	Model	Proven accuracy by [70] ¹¹ or by given citation (%)	Our accuracy (%)
UCM	ResNet-50	98.57	97.77
	ConvNeXt	98.26 [97]	98.84
	SwinT	98.57	97.77
EuroSAT-RGB	ResNet-50	98.83	99.00
	ConvNeXt	98.78	98.34
	SwinT	98.94	98.80
EuroSAT-MS	ResNet-50	96.43 [53]	96.73
	ConvNeXt	<i>No existing work to our knowledge.</i>	97.75
	SwinT		97.34
PatternNet	ResNet-50	99.74	99.52
	ConvNeXt	99.67	99.08
	SwinT	99.69	99.59

department. To maximise usage of available GPU memory, we use a batch size of 64/48/32 when training the ResNet/ConvNeXt/SwinT models respectively. Other training parameters are shown in Table 1. Since our selected datasets have well balanced classes (a similar number of samples per class), we simply report the overall accuracy (accuracy = $\frac{\text{correct predictions}}{\text{total predictions}}$) of our models, achieving performance on par with the state of the art as shown in Table 2.

Having fully trained all models, for each in turn we then proceed to generate explanations via PartitionSHAP, Grad-CAM, and KPCA-CAM (as shown in Figure 1 and Figure 9) for 32 randomly selected samples from each class of each test dataset. We then evaluate the six metrics described in Section 3.4.3 for these samples, again following the configuration detailed in Table 1. Most notably in our configuration, for all datasets apart from EuroSAT-MS, we adopt the pixel shuffling deletion method (randomly rearranging pixels within the desired region) as this is empirically observed to strike a good balance between allowing random perturbations to affect model output and qualitatively maintaining overall image structure to attempt to avoid creating out of domain samples as discussed in footnote 7. For EuroSAT-MS, all three models are incredibly sensitive to shuffling-based deletions¹⁰ resulting in evaluation metrics showing little difference between targeted and randomised approaches, with both slashing model confidence even when just a small proportion of pixels are affected. To effectively evaluate the xAI methods on this dataset therefore, we use the blurring deletion method instead. To generate adversarial examples for evaluating CONTRASTIVITY, we use the DeepFool ℓ_∞ method [94], as implemented by [95], [96], with perturbation degree limiting $\epsilon = 0.05$ for EuroSAT-MS and $\epsilon = 0.01$ for all other datasets.

10. We theorise this is because MS data is more often discontinuous (see Figure 3(d)) and so shuffling leads much more quickly to out of domain samples. The lower comparative resolution of some bands (even after upscaling) potentially exacerbates this issue.

11. These authors use ConvNeXt-Tiny, the variant one step smaller than the Small variant we use with approximately 20 million fewer parameters.

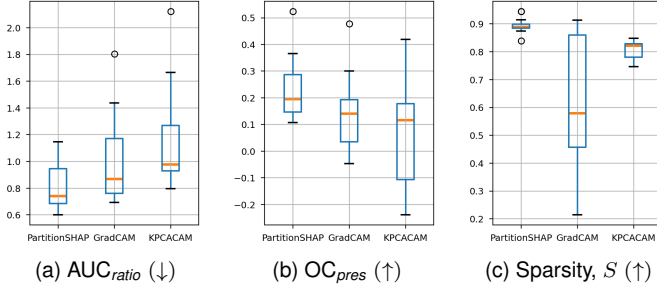


Fig. 8. Box plots of selected metrics, each considering 12 data points per xAI method. The central orange line is the median and the box extends to the first and third quartiles. Whiskers extend to the furthest point within $1.5 \times$ the interquartile range. Flier points are circles past the whiskers.

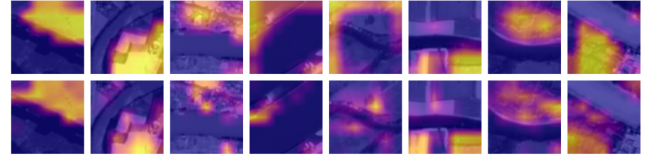
4.2 Metric-based Evaluation of Explanations

It should first be noted, as by [1], that it is unlikely that a particular explanation method scores highly across every single property/metric—some metrics are also partially mutually exclusive—so it is best to view explanations holistically, selecting an explanation which meets the needs of a particular use case. That being said, we proceed to examine the properties in turn. A highly generalised summary of our evaluation results, presenting the mean for each item across all twelve combinations, can be found in Table 3 and selected data are visualised in more detail via box plots in Figure 8. Full data, up to the level of individual class labels, are included in `xlsx` format in the accompanying source code and online.

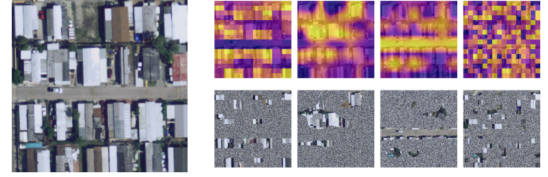
The model randomisation test acts primarily as a sanity check for CORRECTNESS so ‘optimal’ values vary based on the nature of the explanation itself. That being said, the explanations produced by KPCA-CAM of the trained model are significantly more similar to those of the randomised model by the Spearman’s rank, ρ , and top- m intersection, I_m , metrics when compared to the other two methods. This is particularly pronounced for the {EuroSAT-MS, ResNet-50} DMC, with explanations highlighting near identical regions (even if the heatmap intensity is weaker) as shown for the River class in Figure 9(a).

For the AUC_{ratio} metric, plotted in Figure 8(a), only PartitionSHAP performs better than random on average. The CAM methods only outperform randomised explanations in approximately half of DMCs and in some cases perform far worse, with the 1.8+ fliers for both CAM methods corresponding to the {PatternNet, ResNet-50} DMC.

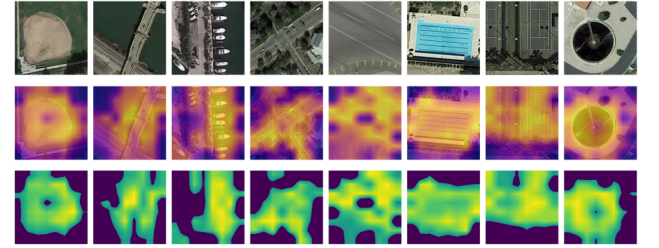
PartitionSHAP is also significantly more successful in identifying important pixels when measured by metrics assessing OUTPUT-COMPLETENESS and is always (apart from



(a) KPCA-CAM explanations for a trained {EuroSAT-MS, ResNet-50} DMC (top) are often very similar to a randomised model (bottom).



(b) Preservation check images (bottom) to measure OUTPUT-COMPLETENESS for a `mobilehomepark` sample (left) from the {UCM, SwinT} DMC for explanations from PartitionSHAP, Grad-CAM, KPCA-CAM, and randomised (top). The confidence in the original output class drops from $1.0000 \rightarrow \{0.9864, 0.0102, 0.0005, 0.7030\}$.



(c) Grad-CAM explanations (middle) have poor COMPACTNESS for the {PatternNet, SwinT} DMC. The bottom row visualises the normalised heatmaps, $\bar{H}$, with $S = \{0.41, 0.44, 0.46, 0.46, 0.25, 0.34, 0.29, 0.40\}$.

Fig. 9. All descriptions correspond to the images from left to right. Visualisation and evaluation of selected explanations. In explanations, yellow, lighter regions indicate important pixels; purple, darker regions unimportant pixels.

for the {PatternNet, ResNet-50} DMC) at least better than random ($OC_{pres} > 0$). Again, KPCA-CAM often actually performs *worse* than a random selection of important pixels, especially for SwinT DMCs. It is also surprisingly inconsistent between DMCs, as is clear from the whisker extent in Figure 8(b). A comparison of the efficacy of relevant pixel preservation of explanations is given in Figure 9(b) (a smaller drop is better).

In terms of CONTRASTIVITY, both Grad-CAM and PartitionSHAP’s explanations reassuringly change markedly from their original form for adversarial inputs across all three similarity metrics. Explanations generated

TABLE 3
Summary of Evaluation Results (Mean)

xAI Method	COR – RT ^a			COR – AUC_{ratio} (↓)	OC_{del} (↑)	OC_{pres} (↑)	CON – TS (↓) ^b			COM – S (↑) ^c
	ρ	I_m	SSIM				ρ	I_m	SSIM	
PartitionSHAP	0.027	0.146	0.423	0.813	0.113	0.233	0.090	0.198	0.450	0.891
Grad-CAM	0.062	0.120	0.343	1.074	0.013	0.145	0.315	0.194	0.506	0.590
KPCA-CAM	0.129	0.186	0.346	1.240	−0.028	0.047	0.641	0.627	0.627	0.797

Best in **bold**. ^a CORRECTNESS – Randomisation Test ^b CONTRASTIVITY – Target Sensitivity ^c COMPACTNESS – Sparsity

via KPCA-CAM however, change far less than we would expect, with on average 62.7% of pixels shared between considered explanations and a very high average SSIM value too (0.627): KPCA-CAM does not appear to be very class sensitive, producing similar explanations for similar inputs even if the model output has changed completely.

Finally examining COMPACTNESS, both PartitionSHAP and KPCA-CAM consistently produce very compact explanations across DMCs, highlighting only approximately up to 10-25% of image pixels as important as can be seen plotted in Figure 8(c). However, Grad-CAM demonstrates huge variability in its COMPACTNESS with some explanations almost uniform ($S = 0.2-0.3$) for SwinT DMCs as exemplified in Figure 9(c) – such explanations provide little actionable insight into a model’s decision-making process. Grad-CAM’s explanations for ResNet-50 DMCs also have lower S around 0.5-0.6.

4.3 Observations

Across all metrics, KPCA-CAM performs poorly, with a consistent pattern of explanations which do not reflect underlying model behaviour. This is especially evident in explanations’ poor CONTRASTIVITY. This could very likely be due to the lack of gradient information in its formulation inhibiting its class discriminative ability, which the original authors also noted in their comparisons with other CAM methods via the RemOve And Debias (ROAD) metric [86]. Explanations frequently merely look like edge detectors or colour segmentation maps, especially unhelpful for remote sensing images which may lack a clearly defined object. The originally proposed evaluations on the ImageNet dataset which showed “more precise activation maps” [12] may simply have been a product of more clearly defined single object images rather than being reflective of the method’s capability to explain the underlying models; that is to say, the intersection over union (IoU) metric for COHERENCE was indicative of good object localisation but not necessarily faithful explanations.

Separately, while they may lack flexibility to the remote sensing domain specifically, across all metrics no methods show significantly worse performance on ConvNeXt or SwinT in comparison to ResNet-50. This suggests all three explanation methods can be reasonably applied to new design paradigms and are flexible to new building block combinations. Reassuringly, neither does the deepness of these two newer networks appear to affect explanation quality. Considered conversely however, the poor relative performance on ResNet-50 is also surprising, especially for Grad-CAM, since this is an architecture these methods were originally designed for and tested on. The only exception is in Grad-CAM’s even lower COMPACTNESS scores for SwinT DMCs, likely again a result of the evaluation domain with large areas of similar token activity a consequence of classification by texture/region-information rather than via the localised feature information considered by the other two more traditional convolution-based architectures. PartitionSHAP seems to avoid these issues due to its model-agnostic consideration of every pixel region separately rather than as part of a singular feature map from within the model itself.

5 EVALUATION

5.1 Strengths and Limitations of Selected Evaluation Metrics

By considering six different metrics, and multiple within each property, for the evaluation of each explanation we were able to build a holistic, relative picture of each method compared to if we had only considered one or two as is common [1]. In doing so we were able to provide strong quantitative evidence for any qualitative observations. This also applies to the measures of similarity we employed – just one measure would not have provided the full picture, especially evident for the randomisation test for CORRECTNESS for example. Our implementation also has the advantage of having given ideal scores for each metric, enabling the targetted optimisation of explanations in future work. Working with any PyTorch model and any explanation method assigning importance values to different pixels, the metric implementations are also tunable and generalisable, enabling their application to wider variety of domains beyond this work.

However, we must also consider some of the limitations of these metrics, although these are sometimes domain-specific. The primary concern is as discussed in footnote 7 and exemplified in Figure 9(b): pixel deletion methods can quickly lead to out of domain samples, potentially invalidating the use of the original model [85]. It is possible PartitionSHAP only scored highly in metrics utilising such deletion because its important pixels were often more evenly distributed throughout the image and therefore less likely to change the prediction class, rather than because their significance to the model was actually higher than those proposed by the CAM methods. This could explain why all models performed best on SwinT for the AUC_{ratio} metric since the transformer image token-based structure may be more robust to deletion perturbations localised to just a few 8×8 pixel windows but not to ones distributed throughout the image affecting many image tokens, as in the random explanations.

This work also did not investigate the impact of differing methods of generating such random explanations. The grid based solution used in this work does not produce ‘realistic’ explanations – an approach based on assigning super-pixel regions random importance values could have provided a better evaluation baseline.

Considering COMPACTNESS specifically, as alluded to at the end of Section 4.3, it could be argued that explanations for LULC images should *not* be unconditionally compact at all, given there may not be a distinct object in the scene. However, this also raises the argument that existing explanation methods focused on saliency heatmaps highlighting particular regions may not be suitable to LULC problems in the first place.

Similarly, none of the explanation methods separately considered the rich multi-channel nature of the input to the MS models, relying only on images’ spatial information rather than this dimensional aspect. Empirical evaluations such as in [16], [53] have concretely demonstrated the varying importance of different bands and their combinations to deep models, all of which are not accounted for here. Likewise, the simplification of incredibly complex models to

a heatmap representing a single layer is potentially hugely reductive and it may be more insightful into full model behaviour to find a way to visualise relations between layers themselves, building up to ultimate classification. This is especially the case for CAM methods, such as those considered, which are originally of a much lower resolution and are scaled up for visualisation: the original explanation may only have a resolution of 7×7 . This may have been additionally compounded in this work by the significant upscaling of EuroSAT images for classification.

5.2 Metrics Not Considered

There were also several areas where more detailed evaluation could have been conducted but was not due to time constraints (see Section 5.4). A successful measure of CONTINUITY (investigating how much explanations change in response to small perturbations in the input), via e.g. **Max Sensitivity** [65], would have been particularly interesting given the huge potential variation in remote sensing imagery due to the large number of uncontrollable environmental variables in deployment contexts. The continuity of the regions themselves, whether they are made up of one contiguous region or many smaller disparate ones, could also have been examined. From our own experiments, this was mainly an issue for PartitionSHAP, with the way in which it considers every pixel region leading to very fine-grained explanations, as is evident comparing the distribution within the image of its top 20% pixels in Figure 9(b) with those of the other explanations which are more clustered. The max evaluations parameter, l , could have been tuned to reduce this problem but, similarly to the above discussion, the extent to which explanations for LULC remote sensing models should even conform to such a metric is debatable given the potential prevalence of large uniform regions within images. A more targetted tuning of l could also have helped address the time disparity in generating explanations – the time taken to generate PartitionSHAP explanations (e.g. over thirty minutes for 32 PatternNet images and SwinT) was prohibitive on the number of experiments that could be run. Such explanations would only be useful for analysis during model development to hone concept learning rather than for end users.

5.3 Response to Research Question

Returning to our original research question, we have demonstrated that the extent to which xAI techniques can be applied to deep learning models for remote sensing LULC classification can depend significantly on the exact dataset and model combination being considered. CAM-based methods especially can struggle with localising important regions for LULC imagery, resulting in uniform, uninformative explanations. In a wider context, we have also shown that convincing explanations are not necessarily reflective of models’ inner workings and that special care should be taken to probe the target sensitivity of explanations to ensure that a model *decision* and not just an image itself is being explained. Informed by these metrics, we have discussed how some types of explanations may not be suitable to remote sensing and found that the evaluation of xAI methods applied to this domain’s images can be

particularly challenging, notably when it comes to deleting and preserving regions according to assigned importance, a strategy which many current evaluation metrics rely upon. We also believe that the configuration of certain metrics, most obviously COMPACTNESS in this work, needs to be standardised and tailored to the image domain under consideration so that different explanation methods can be robustly compared: parameters (e.g. τ) should therefore ideally be determined via user-studies in order to meet the specific needs of domain end users.

5.4 Evaluation of Project Organisation, Approach, and Software Solution

Due to the nature of this as a self-proposed project, we faced some difficulties initially deciding upon specific project objectives and milestones, slowing progress at first. While development during this time on the dataset loading and model training frameworks was helpful, large portions ended up being rewritten from scratch for the final project implementation. A further exploration of semi- and unsupervised techniques ultimately fell out of scope and was also not included in this work. A more focused and detailed literature survey working from seminal and review papers from the outset would have helped to give the project clearer direction from the start, reducing time lost to scope creep, and sped up the identification of a suitable literature gap. This process also meant that the deliverables and milestone timelines initially proposed in October were *not* met, although the overarching research question did remain the same.

However, once new objectives and the final clear project vision had been concretely established in January in terms of an evaluation of xAI methods on a set product of datasets and models, progress was much faster and measurable thanks to clear week by week goals, now able to be set together with our supervisor. Overall, this all effectively resulted in the project’s development progress being shifted later, increasing work load over the Easter period which would have been preferably avoided. That being said, we do not believe that the quality of the project was adversely affected and the management of progress towards objectives, once firmly established, was sound. Several implementation decisions early during development, including the use of Weights and Biases for monitoring model training and the inclusion of a logging system to monitor execution across project files, significantly eased debugging and contributed to timely completion. The final software solution is easily extendable thanks to its object oriented approach and, spurred by the potentially long time taken to generate explanations, includes robust checkpoint resumption/state saving methods both for model training and to load previously computed explanations.

In terms of implementation issues, we were unable to successfully develop a multi-threaded version of PartitionSHAP meaning that the execution time of this explanation method was a limiting factor in the number of experiments we were able to run. This meant that we were also unable to calculate metrics for a ResNet-101 model (to enable a direct comparison of the effect of network depth) or consider a different Sentinel-2 style satellite imagery dataset (e.g. Big

Earth Net) as initially hoped. Logging helped us to identify that our initial implementation of our own CONTINUITY metric was also flawed in that it led to the model changing its predicted class too often (invalidating the comparison of resultant explanations) meaning we were not able to include any of its results in the final report.

More conceptually, the solely individual development of this work has obviously meant a lack of input from the wider xAI field or those with existing experience. For this reason, to ensure the robust reproduction and interrogation of results, the inclusion of full training and evaluation code and chosen parameters is key to this work. Comparison with existing work would have been ideal but unfortunately the primary reference work by [62] does not examine the same datasets so a direct comparison and secondary verification of the results of this work was not possible. Many previous papers also do not provide an implementation of their methods so all metrics were implemented from scratch, again making comparison difficult.

6 CONCLUSION

Taken together, our evaluation results and discussions show that there are no shortcuts to reliable explainability: the notion of a ‘good’ explanation and xAI method is a complex, multi-faceted concept which is *not* domain-agnostic. However, our metrics do show that some existing xAI methods, notably KPCA-CAM, can produce deceptive explanations, at least when applied outside of their original target domain, emphasising that care should be taken to select xAI methods for each particular use case and tailored to the needs of end users. It also highlights that examining just a few properties, only COHERENCE and CORRECTNESS in the case of KPCA-CAM [12] (metric classification ours), is insufficient to confidently establish the efficacy of such methods. Equally, our results have shown that popular methods in the domain, in Grad-CAM and SHAP, do perform well but that their application should still be justified and tuned for the task under consideration. Reaffirming the findings of [1], [9], we have demonstrated that a quantitative assessment of these qualities is possible and provided an extensible framework by which such evaluations can be more easily undertaken by future work. However, following our application of these metrics to the remote sensing LULC problem, we have also discussed their flaws, the need for domain-specific adaptations to be developed, and the demand for standardisation to enable robust comparison of future proposed xAI methods.

As touched on several times by this work, a stand-alone investigation of the impact of using different deletion methods for the evaluation of explanations specifically in the remote sensing domain is still needed, especially for MS data for which models appear to be incredibly sensitive to perturbations. Methods to adequately provide explanations for more complex models utilising band combinations (e.g. [74]) should also be explored by future work. While we have focused on largely model-agnostic xAI methods applicable to both transformers and CNNs, the recent explosion in transformer-based architectures specifically brings with it a need for architecture-specific explanations such as visualising attention weights [98] alongside appropriate new

metrics which can be used to evaluate their effectiveness. Apart from also considering further properties not evaluated in this work including CONTINUITY and COHERENCE [1] (utilising datasets providing segmentation maps, e.g. [20], [21]), future work could additionally investigate the extent to which explanations and their faithfulness to model behaviour differ for *incorrect* model outputs specifically, a distinction not made by this work.

REFERENCES

- [1] M. Nauta, J. Trienes, S. Pathak, E. Nguyen, M. Peters, Y. Schmitt, J. Schlötterer, M. van Keulen, and C. Seifert, "From anecdotal evidence to quantitative evaluation methods: A systematic review on evaluating explainable ai," *ACM Comput. Surv.*, vol. 55, no. 13s, Jul. 2023. [Online]. Available: <https://doi.org/10.1145/3583558>
- [2] sp@ceboy. (2021) Orbiting now – active satellite tle data and information. Accessed: Apr. 16, 2025. [Online]. Available: <https://orbiting-now.com/>
- [3] D. Mohney. (2020) Terabytes from space: Satellite imaging is filling data centers. Accessed: Apr. 16, 2025. [Online]. Available: <https://www.datacenterfrontier.com/internet-of-things/article/11429032/terabytes-from-space-satellite-imaging-is-filling-data-centers>
- [4] G. Cheng, X. Xie, J. Han, L. Guo, and G.-S. Xia, "Remote sensing image scene classification meets deep learning: Challenges, methods, benchmarks, and opportunities," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 13, pp. 3735–3756, 2020. [Online]. Available: <https://doi.org/10.1109/JSTARS.2020.3005403>
- [5] Copernicus. (2025) S2 mission. Accessed: Apr. 16, 2025. [Online]. Available: <https://sentiwiki.copernicus.eu/web/s2-mission>
- [6] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "Imagenet classification with deep convolutional neural networks," in *Advances in Neural Information Processing Systems*, F. Pereira, C. Burges, L. Bottou, and K. Weinberger, Eds., vol. 25. Curran Associates, Inc., 2012. [Online]. Available: <https://proceedings.neurips.cc/paper/2012/hash/c399862d3b9d6b76c8436e924a68c45b-Abstract.html>
- [7] O. Russakovsky, J. Deng, H. Su, J. Krause, S. Satheesh, S. Ma, Z. Huang, A. Karpathy, A. Khosla, M. Bernstein, A. C. Berg, and L. Fei-Fei, "Imagenet large scale visual recognition challenge," *International Journal of Computer Vision*, vol. 115, no. 3, pp. 211–252, 2015. [Online]. Available: <https://doi.org/10.1007/s11263-015-0816-y>
- [8] J. Wentworth and B. Esen. (2020) Remote sensing and machine learning. Accessed: Apr. 16, 2025. [Online]. Available: <https://post.parliament.uk/research-briefings/post-pn-0628/>
- [9] A. Höhl, I. Obadic, M.-A. Fernández-Torres, H. Najjar, D. A. B. Oliveira, Z. Akata, A. Dengel, and X. X. Zhu, "Opening the black box: A systematic review on explainable artificial intelligence in remote sensing," *IEEE Geoscience and Remote Sensing Magazine*, vol. 12, no. 4, pp. 261–304, 2024. [Online]. Available: <https://doi.org/10.1109/MGRS.2024.3467001>
- [10] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems*, I. Guyon, U. V. Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, Eds., vol. 30. Curran Associates, Inc., 2017. [Online]. Available: <https://proceedings.neurips.cc/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html>
- [11] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-cam: Visual explanations from deep networks via gradient-based localization," in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, Oct 2017. [Online]. Available: https://openaccess.thecvf.com/content_iccv_2017/html/Selvaraju_Grad-CAM_Visual_Explanations_ICCV_2017_paper.html
- [12] S. Karmani, T. Sivakaran, G. Prasad, M. Ali, W. Yang, and S. Tang, "Kpca-cam: Visual explainability of deep computer vision models using kernel pca," in *2024 IEEE 26th International Workshop on Multimedia Signal Processing (MMSP)*, 2024, pp. 1–5. [Online]. Available: <https://doi.org/10.1109/MMSP61759.2024.10743968>
- [13] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2016. [Online]. Available: https://openaccess.thecvf.com/content_cvpr_2016/html/He_Deep_Residual_Learning_CVPR_2016_paper.html
- [14] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A convnet for the 2020s," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2022, pp. 11 976–11 986. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2022/html/Liu_A_ConvNet_for_the_2020s_CVPR_2022_paper.html
- [15] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo, "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, October 2021, pp. 10 012–10 022. [Online]. Available: https://openaccess.thecvf.com/content/ICCV2021/html/Liu_Swin_Transformer_Hierarchical_Vision_Transformer_Using_Shifted_Windows_ICCV_2021_paper
- [16] P. Helber, B. Bischke, A. Dengel, and D. Borth, "Eurosat: A novel dataset and deep learning benchmark for land use and land cover classification," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 12, no. 7, pp. 2217–2226, 2019. [Online]. Available: <https://doi.org/10.1109/JSTARS.2019.2918242>
- [17] Y. Yang and S. Newsam, "Bag-of-visual-words and spatial extensions for land-use classification," in *Proceedings of the 18th SIGSPATIAL International Conference on Advances in Geographic Information Systems*, ser. GIS '10. New York, NY, USA: Association for Computing Machinery, 2010, p. 270–279. [Online]. Available: <https://doi.org/10.1145/1869790.1869829>
- [18] W. Zhou, S. Newsam, C. Li, and Z. Shao, "Patternnet: A benchmark dataset for performance evaluation of remote sensing image retrieval," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 145, pp. 197–209, 2018, deep Learning RS Data. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0924271618300042>
- [19] S. Basu, S. Ganguly, S. Mukhopadhyay, R. DiBiano, M. Karki, and R. Nemani, "Deepsat: a learning framework for satellite imagery," in *Proceedings*

- of the 23rd SIGSPATIAL International Conference on Advances in Geographic Information Systems, ser. SIGSPATIAL '15. New York, NY, USA: Association for Computing Machinery, 2015. [Online]. Available: <https://doi.org/10.1145/2820783.2820816>
- [20] G. Sumbul, M. Charfuelan, B. Demir, and V. Markl, "Bigearthnet: A large-scale benchmark archive for remote sensing image understanding," in *IGARSS 2019 - 2019 IEEE International Geoscience and Remote Sensing Symposium*, 2019, pp. 5901–5904. [Online]. Available: <https://doi.org/10.1109/IGARSS.2019.8900532>
- [21] A. Garioud, N. Gonthier, L. Landrieu, A. De Wit, M. Valette, M. Poupée, S. Giordano, and b. Wattrelos, "Flair : a country-scale land cover semantic segmentation dataset from multi-source optical imagery," in *Advances in Neural Information Processing Systems*, A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, Eds., vol. 36. Curran Associates, Inc., 2023, pp. 16 456–16 482. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2023/hash/353ca88f722cdd0c481b999428ae113a-Abstract-Datasets_and_Benchmarks.html
- [22] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, "An image is worth 16x16 words: Transformers for image recognition at scale," 2021. [Online]. Available: <https://arxiv.org/abs/2010.11929>
- [23] H. Wang, Z. Wang, M. Du, F. Yang, Z. Zhang, S. Ding, P. Mardziel, and X. Hu, "Score-cam: Score-weighted visual explanations for convolutional neural networks," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, June 2020. [Online]. Available: https://openaccess.thecvf.com/content_CVPRW_2020/html/w1/Wang_Score-CAM_Score-Weighted_Visual_Explanations_for_Convolutional_Neural_Networks_CVPRW_2020_paper.html
- [24] M. B. Muhammad and M. Yeasin, "Eigen-cam: Class activation map using principal components," in *2020 International Joint Conference on Neural Networks (IJCNN)*, 2020, pp. 1–7. [Online]. Available: <https://doi.org/10.1109/IJCNN48605.2020.9206626>
- [25] D. G. Lowe, "Distinctive image features from scale-invariant keypoints," *International Journal of Computer Vision*, vol. 60, no. 2, pp. 91–110, 2004. [Online]. Available: <https://doi.org/10.1023/B:VISI.0000029664.99615.94>
- [26] Y. Yang and S. Newsam, "Comparing sift descriptors and gabor texture features for classification of remote sensed imagery," in *2008 15th IEEE International Conference on Image Processing*, 2008, pp. 1852–1855. [Online]. Available: <https://doi.org/10.1109/ICIP.2008.4712139>
- [27] J. A. dos Santos, O. A. B. Penatti, and R. da S. Torres, "Evaluating the potential of texture and color descriptors for remote sensing image retrieval and classification," in *Proceedings of the International Conference on Computer Vision Theory and Applications (VISIGRAPP 2010) - Volume 2: VISAPP, INSTICC*. SciTePress, 2010, pp. 203–208. [Online]. Available: <https://www.scitepress.org/Papers/2010/28434/>
- [28] P. M. Atkinson and A. R. L. Tatnall, "Introduction neural networks in remote sensing," *International Journal of Remote Sensing*, vol. 18, no. 4, pp. 699–709, 1997. [Online]. Available: <https://doi.org/10.1080/014311697218700>
- [29] K. Fukushima, "Neocognitron: A self-organizing neural network model for a mechanism of pattern recognition unaffected by shift in position," *Biological Cybernetics*, vol. 36, no. 4, pp. 193–202, 1980. [Online]. Available: <https://doi.org/10.1007/BF00344251>
- [30] C. Szegedy, W. Liu, Y. Jia, P. Sermanet, S. Reed, D. Anguelov, D. Erhan, V. Vanhoucke, and A. Rabinovich, "Going deeper with convolutions," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2015. [Online]. Available: https://www.cv-foundation.org/openaccess/content_cvpr_2015/html/Szegedy_Going_Deeper_With_2015_CVPR_paper.html
- [31] L. Ma, Y. Liu, X. Zhang, Y. Ye, G. Yin, and B. A. Johnson, "Deep learning in remote sensing applications: A meta-analysis and review," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 152, pp. 166–177, 2019. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0924271619301108>
- [32] G. J. Scott, M. R. England, W. A. Starns, R. A. Marcum, and C. H. Davis, "Training deep convolutional neural networks for land-cover classification of high-resolution imagery," *IEEE Geoscience and Remote Sensing Letters*, vol. 14, no. 4, pp. 549–553, 2017. [Online]. Available: <https://doi.org/10.1109/LGRS.2017.2657778>
- [33] M. Castelluccio, G. Poggi, C. Sansone, and L. Verdoliva, "Land use classification in remote sensing images by convolutional neural networks," 2015. [Online]. Available: <https://arxiv.org/abs/1508.00092>
- [34] M. Schmitt, L. Hughes, C. Qiu, and X. Zhu, "Sen12ms—a curated dataset of georeferenced multi-spectral sentinel-1/2 imagery for deep learning and data fusion," *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, vol. 4, pp. 153–160, 2019. [Online]. Available: <https://doi.org/10.5194/isprs-annals-IV-2-W7-153-2019>
- [35] M. Neumann, A. S. Pinto, X. Zhai, and N. Houlsby, "In-domain representation learning for remote sensing," 2019. [Online]. Available: <https://arxiv.org/abs/1911.06721>
- [36] S. Jozdani, D. Chen, D. Pouliot, and B. Alan Johnson, "A review and meta-analysis of generative adversarial networks and their applications in remote sensing," *International Journal of Applied Earth Observation and Geoinformation*, vol. 108, p. 102734, 2022. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0303243422000605>
- [37] I. J. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, and Y. Bengio, "Generative adversarial nets," in *Advances in Neural Information Processing Systems*, Z. Ghahramani, M. Welling, C. Cortes, N. Lawrence, and K. Weinberger, Eds., vol. 27.

- Curran Associates, Inc., 2014. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2014/hash/f033ed80deb0234979a61f95710dbe25-Abstract.html
- [38] D. Lin, K. Fu, Y. Wang, G. Xu, and X. Sun, "Marta gans: Unsupervised representation learning for remote sensing image classification," *IEEE Geoscience and Remote Sensing Letters*, vol. 14, no. 11, pp. 2092–2096, 2017. [Online]. Available: <https://doi.org/10.1109/LGRS.2017.2752750>
- [39] Y. Yu, X. Li, and F. Liu, "Attention gans: Unsupervised deep feature learning for aerial scene classification," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 58, no. 1, pp. 519–531, 2020. [Online]. Available: <https://doi.org/10.1109/TGRS.2019.2937830>
- [40] Y. Bazi, L. Bashmal, M. M. A. Rahhal, R. A. Dayil, and N. A. Ajlan, "Vision transformers for remote sensing image classification," *Remote Sensing*, vol. 13, no. 3, 2021. [Online]. Available: <https://www.mdpi.com/2072-4292/13/3/516>
- [41] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. u. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems*, I. Guyon, U. V. Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, Eds., vol. 30. Curran Associates, Inc., 2017. [Online]. Available: <https://proceedings.neurips.cc/paper/2017/hash/3f5ee243547dee91fbd053c1c4a845aa-Abstract.html>
- [42] M. T. Ribeiro, S. Singh, and C. Guestrin, "'Why Should I Trust You?': Explaining the predictions of any classifier," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, ser. KDD '16. New York, NY, USA: Association for Computing Machinery, 2016, p. 1135–1144. [Online]. Available: <https://doi.org/10.1145/2939672.2939778>
- [43] K. Patel, N. Bancroft, S. M. Drucker, J. Fogarty, A. J. Ko, and J. Landay, "Gestalt: integrated support for implementation and analysis in machine learning," in *Proceedings of the 23rd Annual ACM Symposium on User Interface Software and Technology*, ser. UIST '10. New York, NY, USA: Association for Computing Machinery, 2010, p. 37–46. [Online]. Available: <https://doi.org/10.1145/1866029.1866038>
- [44] S. Amershi, M. Chickering, S. M. Drucker, B. Lee, P. Simard, and J. Suh, "Modeltracker: Redesigning performance analysis tools for machine learning," in *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*, ser. CHI '15. New York, NY, USA: Association for Computing Machinery, 2015, p. 337–346. [Online]. Available: <https://doi.org/10.1145/2702123.2702509>
- [45] M. D. Zeiler and R. Fergus, "Visualizing and understanding convolutional networks," in *Computer Vision – ECCV 2014*, D. Fleet, T. Pajdla, B. Schiele, and T. Tuytelaars, Eds. Cham: Springer International Publishing, 2014, pp. 818–833. [Online]. Available: https://doi.org/10.1007/978-3-319-10590-1_53
- [46] K. Simonyan, A. Vedaldi, and A. Zisserman, "Deep inside convolutional networks: Visualising image classification models and saliency maps," 2014. [Online]. Available: <https://arxiv.org/abs/1312.6034>
- [47] A. Vedaldi and S. Soatto, "Quick shift and kernel methods for mode seeking," in *Computer Vision – ECCV 2008*, D. Forsyth, P. Torr, and A. Zisserman, Eds. Berlin, Heidelberg: Springer Berlin Heidelberg, 2008, pp. 705–718. [Online]. Available: https://doi.org/10.1007/978-3-540-88693-8_52
- [48] M. Verma, N. Gupta, B. Tolani, and R. Kaushal, "Explainable custom cnn architecture for land use classification using satellite images," in *2021 Sixth International Conference on Image Information Processing (ICIIP)*, vol. 6, 2021, pp. 304–309. [Online]. Available: <https://doi.org/10.1109/ICIIP53038.2021.9702698>
- [49] A. Shrikumar, P. Greenside, and A. Kundaje, "Learning important features through propagating activation differences," in *Proceedings of the 34th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, D. Precup and Y. W. Teh, Eds., vol. 70. PMLR, 06–11 Aug 2017, pp. 3145–3153. [Online]. Available: <https://proceedings.mlr.press/v70/shrikumar17a.html>
- [50] M. Sundararajan, A. Taly, and Q. Yan, "Axiomatic attribution for deep networks," in *Proceedings of the 34th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, D. Precup and Y. W. Teh, Eds., vol. 70. PMLR, 06–11 Aug 2017, pp. 3319–3328. [Online]. Available: <https://proceedings.mlr.press/v70/sundararajan17a.html>
- [51] A. Mamalakis, E. A. Barnes, and I. Ebert-Uphoff, "Carefully choose the baseline: Lessons learned from applying xai attribution methods for regression tasks in geoscience," *Artificial Intelligence for the Earth Systems*, vol. 2, no. 1, p. e220058, 2023. [Online]. Available: <https://journals.ametsoc.org/view/journals/aies/2/1/AIES-D-22-0058.1.xml>
- [52] L. Shapley, 7. *A Value for n-Person Games. Contributions to the Theory of Games II* (1953) 307–317. Princeton: Princeton University Press, 1997, pp. 69–79. [Online]. Available: <https://doi.org/10.1515/9781400829156-012>
- [53] A. Temenos, N. Temenos, M. Kaselimi, A. Doulamis, and N. Doulamis, "Interpretable deep learning framework for land use and land cover classification in remote sensing using shap," *IEEE Geoscience and Remote Sensing Letters*, vol. 20, pp. 1–5, 2023. [Online]. Available: <https://doi.org/10.1109/LGRS.2023.3251652>
- [54] B. Zhou, A. Khosla, A. Lapedriza, A. Oliva, and A. Torralba, "Learning deep features for discriminative localization," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2016. [Online]. Available: https://openaccess.thecvf.com/content_cvpr_2016/html/Zhou_Learning_Deep_Features_CVPR_2016_paper.html
- [55] O. Sen and H. Y. Keles, "On the evaluation of cnn models in remote-sensing scene classification domain," *PFG – Journal of Photogrammetry, Remote Sensing and Geoinformation Science*, vol. 88, no. 6, pp. 477–492, 2020. [Online]. Available: <https://doi.org/10.1007/s41064-020-00129-6>
- [56] S. Leem and H. Seo, "Attention guided cam: Visual

- explanations of vision transformer guided by self-attention," *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, no. 4, pp. 2956–2964, Mar. 2024. [Online]. Available: <https://ojs.aaai.org/index.php/AAAI/article/view/28077>
- [57] M. Khan, A. Hanan, M. Kenzhebay, M. Gazzea, and R. Arghandeh, "Transformer-based land use and land cover classification with explainability using satellite imagery," *Scientific Reports*, vol. 14, no. 1, p. 16744, 2024. [Online]. Available: <https://doi.org/10.1038/s41598-024-67186-4>
- [58] A. Chattopadhyay, A. Sarkar, P. Howlader, and V. N. Balasubramanian, "Grad-cam++: Generalized gradient-based visual explanations for deep convolutional networks," in *2018 IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2018, pp. 839–847. [Online]. Available: <https://doi.org/10.1109/WACV.2018.00097>
- [59] X. Guo, B. Hou, C. Yang, S. Ma, B. Ren, S. Wang, and L. Jiao, "Visual explanations with detailed spatial information for remote sensing image classification via channel saliency," *International Journal of Applied Earth Observation and Geoinformation*, vol. 118, p. 103244, 2023. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1569843223000663>
- [60] J. Adebayo, J. Gilmer, M. Muelly, I. Goodfellow, M. Hardt, and B. Kim, "Sanity checks for saliency maps," in *Advances in Neural Information Processing Systems*, S. Bengio, H. Wallach, H. Larochelle, K. Grauman, N. Cesa-Bianchi, and R. Garnett, Eds., vol. 31. Curran Associates, Inc., 2018. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2018/hash/294a8ed24b1ad22ec2e7efea049b8737-Abstract.html
- [61] Y. Rong, T. Leemann, T.-T. Nguyen, L. Fiedler, P. Qian, V. Unhelkar, T. Seidel, G. Kasneci, and E. Kasneci, "Towards human-centered explainable ai: A survey of user studies for model explanations," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 46, no. 4, pp. 2104–2122, 2024. [Online]. Available: <https://doi.org/10.1109/TPAMI.2023.3331846>
- [62] I. Kakogeorgiou and K. Karantzas, "Evaluating explainable artificial intelligence methods for multi-label deep learning classification tasks in remote sensing," *International Journal of Applied Earth Observation and Geoinformation*, vol. 103, p. 102520, 2021. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0303243421002270>
- [63] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 2261–2269. [Online]. Available: <https://doi.org/10.1109/CVPR.2017.243>
- [64] W. Samek, A. Binder, G. Montavon, S. Lapuschkin, and K.-R. Müller, "Evaluating the visualization of what a deep neural network has learned," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 28, no. 11, pp. 2660–2673, 2017. [Online]. Available: <https://doi.org/10.1109/TNNLS.2016.2599820>
- [65] C.-K. Yeh, C.-Y. Hsieh, A. Suggala, D. I. Inouye, and P. K. Ravikumar, "On the (in)fidelity and sensitivity of explanations," in *Advances in Neural Information Processing Systems*, H. Wallach, H. Larochelle, A. Beygelzimer, F. d'Alché-Buc, E. Fox, and R. Garnett, Eds., vol. 32. Curran Associates, Inc., 2019. [Online]. Available: <https://proceedings.neurips.cc/paper/2019/hash/a7471fdc77b3435276507cc8f2dc2569-Abstract.html>
- [66] W. Samek, G. Montavon, S. Lapuschkin, C. J. Anders, and K.-R. Müller, "Explaining deep neural networks and beyond: A review of methods and applications," *Proceedings of the IEEE*, vol. 109, no. 3, pp. 247–278, 2021. [Online]. Available: <https://doi.org/10.1109/JPROC.2021.3060483>
- [67] TorchVision maintainers and contributors, "Torchvision: Pytorch's computer vision library," <https://github.com/pytorch/vision>, 2016.
- [68] V. Vryniotis. (2021) How to train state-of-the-art models using torchvision's latest primitives. Accessed: Apr. 16, 2025. [Online]. Available: <https://pytorch.org/blog/how-to-train-state-of-the-art-models-using-torchvision-latest-primitives/>
- [69] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," 2019. [Online]. Available: <https://arxiv.org/abs/1711.05101>
- [70] I. Dimitrovski, I. Kitanovski, D. Kocev, and N. Simidjievski, "Current trends in deep learning for earth observation: An open-source benchmark arena for image classification," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 197, pp. 18–35, 2023. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0924271623000205>
- [71] F.-E. Jannat and A. R. Willis, "Improving classification of remotely sensed images with the swin transformer," in *SoutheastCon 2022*, 2022, pp. 611–618. [Online]. Available: <https://doi.org/10.1109/SoutheastCon48659.2022.9764016>
- [72] Z. Liu, H. Hu, Y. Lin, Z. Yao, Z. Xie, Y. Wei, J. Ning, Y. Cao, Z. Zhang, L. Dong, F. Wei, and B. Guo, "Swin transformer v2: Scaling up capacity and resolution," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2022, pp. 12 009–12 019. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2022/html/Liu_Swin_Transformer_V2_Scaling_Up_Capacity_and_Resolution_CVPR_2022_paper.html
- [73] A. J. Stewart, C. Robinson, I. A. Corley, A. Ortiz, J. M. Lavista Ferres, and A. Banerjee, "TorchGeo: Deep learning with geospatial data," *ACM Transactions on Spatial Algorithms and Systems*, Dec. 2024. [Online]. Available: <https://doi.org/10.1145/3707459>
- [74] B. Engelbrecht and T. L. van Zyl, "Comparing cnn architectures for land cover classification on multispectral images," in *2021 IEEE International Geoscience and Remote Sensing Symposium IGARSS*, 2021, pp. 5378–5381. [Online]. Available: <https://doi.org/10.1109/IGARSS47720.2021.9553337>
- [75] Y. Sharma and R. Ross, "Less is more when applying transfer learning to multi-spectral data." in *AICS*, 2020, pp. 301–312. [Online].

- Available: https://ceurspt.wikidata.dbis.rwth-aachen.de/Vol-2771/AICS2020_paper_50.pdf
- [76] C. Molnar, *Interpretable Machine Learning*, 3rd ed., 2025. [Online]. Available: <https://christophm.github.io/interpretable-ml-book>
- [77] J. Gildenblat and contributors, "Pytorch library for cam methods," <https://github.com/jacobgil/pytorch-grad-cam>, 2021.
- [78] B. Schölkopf, A. Smola, and K.-R. Müller, "Kernel principal component analysis," in *Artificial Neural Networks — ICANN'97*, W. Gerstner, A. Germond, M. Hasler, and J.-D. Nicoud, Eds. Berlin, Heidelberg: Springer Berlin Heidelberg, 1997, pp. 583–588. [Online]. Available: <https://doi.org/10.1007/BFb0020217>
- [79] R. Tomsett, D. Harborne, S. Chakraborty, P. Gurram, and A. Preece, "Sanity checks for saliency metrics," *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 34, no. 04, pp. 6021–6029, Apr. 2020. [Online]. Available: <https://ojs.aaai.org/index.php/AAAI/article/view/6064>
- [80] Z. Wang, A. Bovik, H. Sheikh, and E. Simoncelli, "Image quality assessment: from error visibility to structural similarity," *IEEE Transactions on Image Processing*, vol. 13, no. 4, pp. 600–612, 2004. [Online]. Available: <https://doi.org/10.1109/TIP.2003.819861>
- [81] Z. Wang and A. C. Bovik, "Mean squared error: Love it or leave it? a new look at signal fidelity measures," *IEEE Signal Processing Magazine*, vol. 26, no. 1, pp. 98–117, 2009. [Online]. Available: <https://doi.org/10.1109/MSP.2008.930649>
- [82] A. Ghorbani, A. Abid, and J. Zou, "Interpretation of neural networks is fragile," *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 33, no. 01, pp. 3681–3688, Jul. 2019. [Online]. Available: <https://ojs.aaai.org/index.php/AAAI/article/view/4252>
- [83] W. J. Murdoch, C. Singh, K. Kumbier, R. Abbasi-Asl, and B. Yu, "Definitions, methods, and applications in interpretable machine learning," *Proceedings of the National Academy of Sciences*, vol. 116, no. 44, pp. 22 071–22 080, 2019. [Online]. Available: <https://www.pnas.org/doi/abs/10.1073/pnas.1900654116>
- [84] V. Petsiuk, A. Das, and K. Saenko, "Rise: Randomized input sampling for explanation of black-box models," 2018. [Online]. Available: <https://arxiv.org/abs/1806.07421>
- [85] S. Hooker, D. Erhan, P.-J. Kindermans, and B. Kim, "A benchmark for interpretability methods in deep neural networks," in *Advances in Neural Information Processing Systems*, H. Wallach, H. Larochelle, A. Beygelzimer, F. d'Alché-Buc, E. Fox, and R. Garnett, Eds., vol. 32. Curran Associates, Inc., 2019. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2019/hash/fe4b855600d0f0cae99daa5c5c5a410-Abstract.html
- [86] Y. Rong, T. Leemann, V. Borisov, G. Kasneci, and E. Kasneci, "A consistent and efficient evaluation strategy for attribution methods," in *Proceedings of the 39th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, K. Chaudhuri, S. Jegelka, L. Song, C. Szepesvari, G. Niu, and S. Sabato, Eds., vol. 162. PMLR, 17–23 Jul 2022, pp. 18 770–18 795. [Online]. Available: <https://proceedings.mlr.press/v162/rong22a.html>
- [87] J. Wagner, J. M. Kohler, T. Gindele, L. Hetzel, J. T. Wiedemer, and S. Behnke, "Interpretable and fine-grained visual explanations for convolutional neural networks," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2019. [Online]. Available: https://openaccess.thecvf.com/content_CVPR_2019/html/Wagner_Interpretable_and_Fine-Grained_Visual_Explanations_for_Convolutional_Neural_Networks_CVPR_2019_paper.html
- [88] T. Miller, "Explanation in artificial intelligence: Insights from the social sciences," *Artificial Intelligence*, vol. 267, pp. 1–38, 2019. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0004370218305988>
- [89] W. Nie, Y. Zhang, and A. Patel, "A theoretical explanation for perplexing behaviors of backpropagation-based visualizations," in *Proceedings of the 35th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, J. Dy and A. Krause, Eds., vol. 80. PMLR, 10–15 Jul 2018, pp. 3809–3818. [Online]. Available: <https://proceedings.mlr.press/v80/nie18a.html>
- [90] I. J. Goodfellow, J. Shlens, and C. Szegedy, "Explaining and harnessing adversarial examples," 2015. [Online]. Available: <https://arxiv.org/abs/1412.6572>
- [91] X. Cui, J. M. Lee, and J. Hsieh, "An integrative 3c evaluation framework for explainable artificial intelligence," in *AMCIS 2019 Proceedings*, 2019. [Online]. Available: https://aisel.aisnet.org/amcis2019/ai_semantic_for_intelligent_info_systems/ai_semantic_for_intelligent_info_systems/10/
- [92] U. Bhatt, A. Weller, and J. M. F. Moura, "Evaluating and aggregating feature-based model explanations," in *Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence, IJCAI-20*, C. Bessiere, Ed. International Joint Conferences on Artificial Intelligence Organization, 7 2020, pp. 3016–3022, main track. [Online]. Available: <https://doi.org/10.24963/ijcai.2020/417>
- [93] J. Ansel, E. Yang, H. He, N. Gimelshein, A. Jain, M. Voznesensky, B. Bao, P. Bell, D. Berard, E. Burovski, G. Chauhan, A. Chourdia, W. Constable, A. Desmaison, Z. DeVito, E. Ellison, W. Feng, J. Gong, M. Gschwind, B. Hirsh, S. Huang, K. Kalambarkar, L. Kirsch, M. Lazos, M. Lezcano, Y. Liang, J. Liang, Y. Lu, C. Luk, B. Maher, Y. Pan, C. Puhersch, M. Reso, M. Saroufim, M. Y. Siraichi, H. Suk, M. Suo, P. Tillet, E. Wang, X. Wang, W. Wen, S. Zhang, X. Zhao, K. Zhou, R. Zou, A. Mathews, G. Chanan, P. Wu, and S. Chintala, "PyTorch 2: Faster Machine Learning Through Dynamic Python Bytecode Transformation and Graph Compilation," in *29th ACM International Conference on Architectural Support for Programming Languages and Operating Systems, Volume 2 (ASPLOS '24)*. ACM, Apr. 2024. [Online]. Available: <https://pytorch.org/assets/pytorch2-2.pdf>
- [94] S.-M. Moosavi-Dezfooli, A. Fawzi, and P. Frossard, "Deepfool: A simple and accurate method to fool deep

- neural networks,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2016. [Online]. Available: https://openaccess.thecvf.com/content_cvpr_2016/html/Moosavi-Dezfooli_DeepFool_A_Simple_CVPR_2016_paper.html
- [95] J. Rauber, R. Zimmermann, M. Bethge, and W. Brendel, “Foolbox native: Fast adversarial attacks to benchmark the robustness of machine learning models in pytorch, tensorflow, and jax,” *Journal of Open Source Software*, vol. 5, no. 53, p. 2607, 2020. [Online]. Available: <https://doi.org/10.21105/joss.02607>
- [96] J. Rauber, W. Brendel, and M. Bethge, “Foolbox: A python toolbox to benchmark the robustness of machine learning models,” in *Reliable Machine Learning in the Wild Workshop, 34th International Conference on Machine Learning*, 2017. [Online]. Available: <http://arxiv.org/abs/1707.04131>
- [97] M. Yang, Y. Chen, X. Lu, and S. Xiong, “Preferred feature representation for remote sensing scene classification,” in *Proceedings of the 2024 16th International Conference on Machine Learning and Computing*, ser. ICMLC '24. New York, NY, USA: Association for Computing Machinery, 2024, p. 418–423. [Online]. Available: <https://doi.org/10.1145/3651671.3651697>
- [98] S. Chaudhari, V. Mithal, G. Polatkan, and R. Ramanath, “An attentive survey of attention models,” *ACM Trans. Intell. Syst. Technol.*, vol. 12, no. 5, Oct. 2021. [Online]. Available: <https://doi.org/10.1145/3465055>